Encontro Nacional de Física Estatística - ENFE
November 07, Curitiba, Paraná, Brasil

Analysis of Socioeconomic Indicators in Brazil, USA and others Latin Americas Countries using Econophysics Techniques

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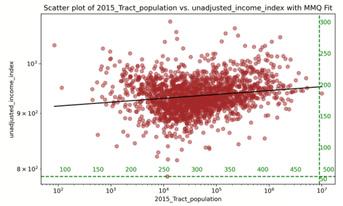
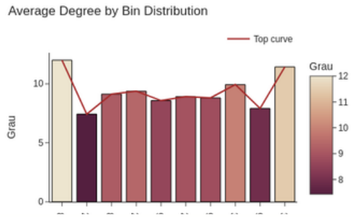
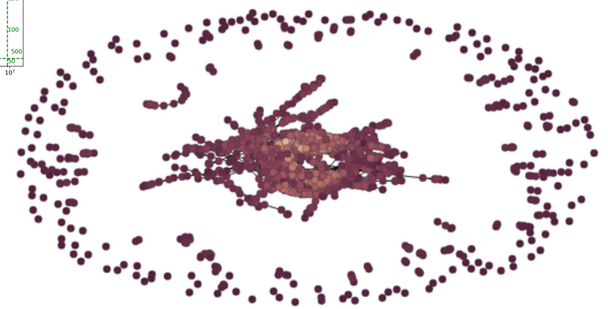
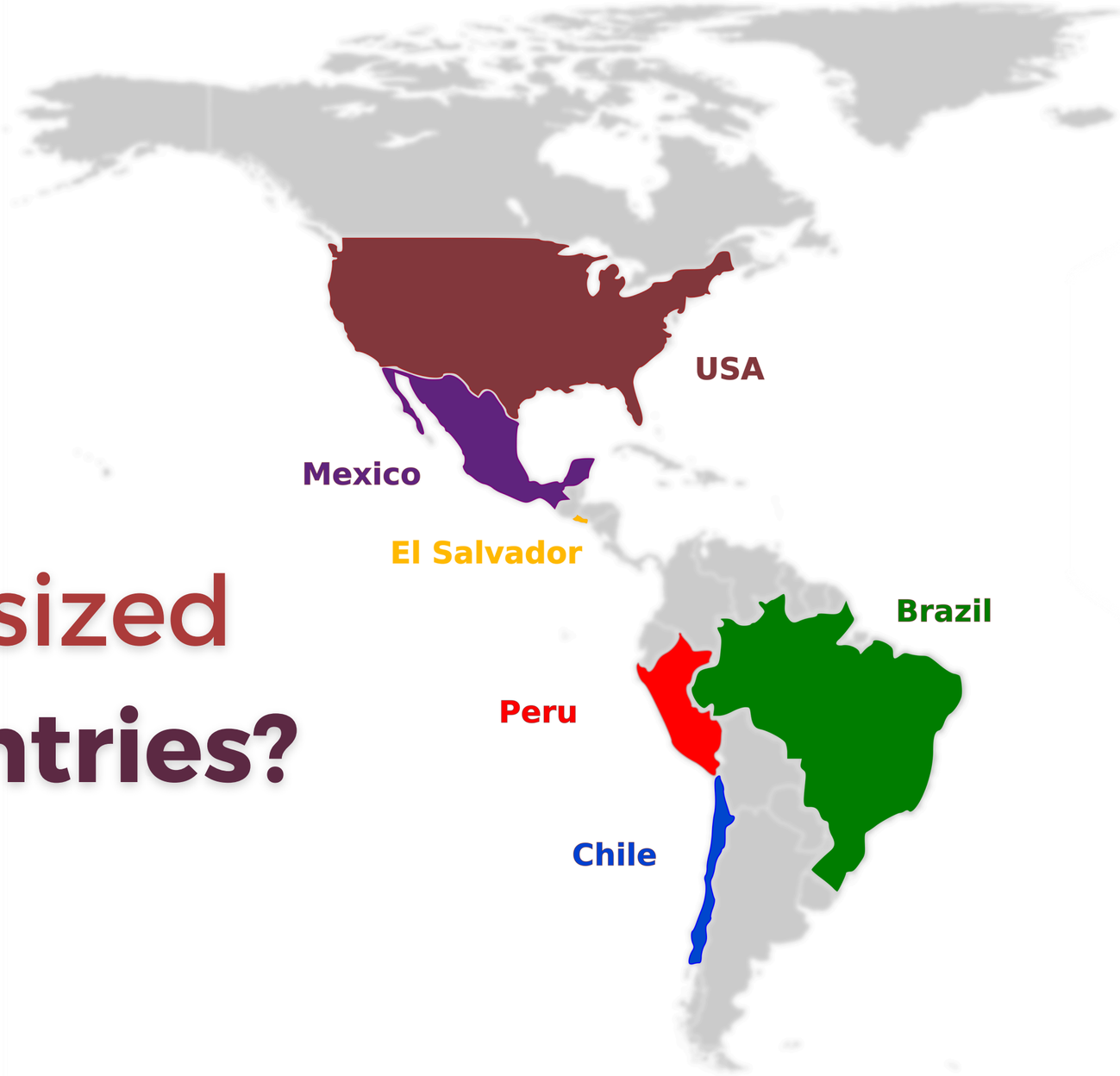






This work represents the overview about my master's and future Phd theses. *

How to compare small and medium-sized cities between countries?



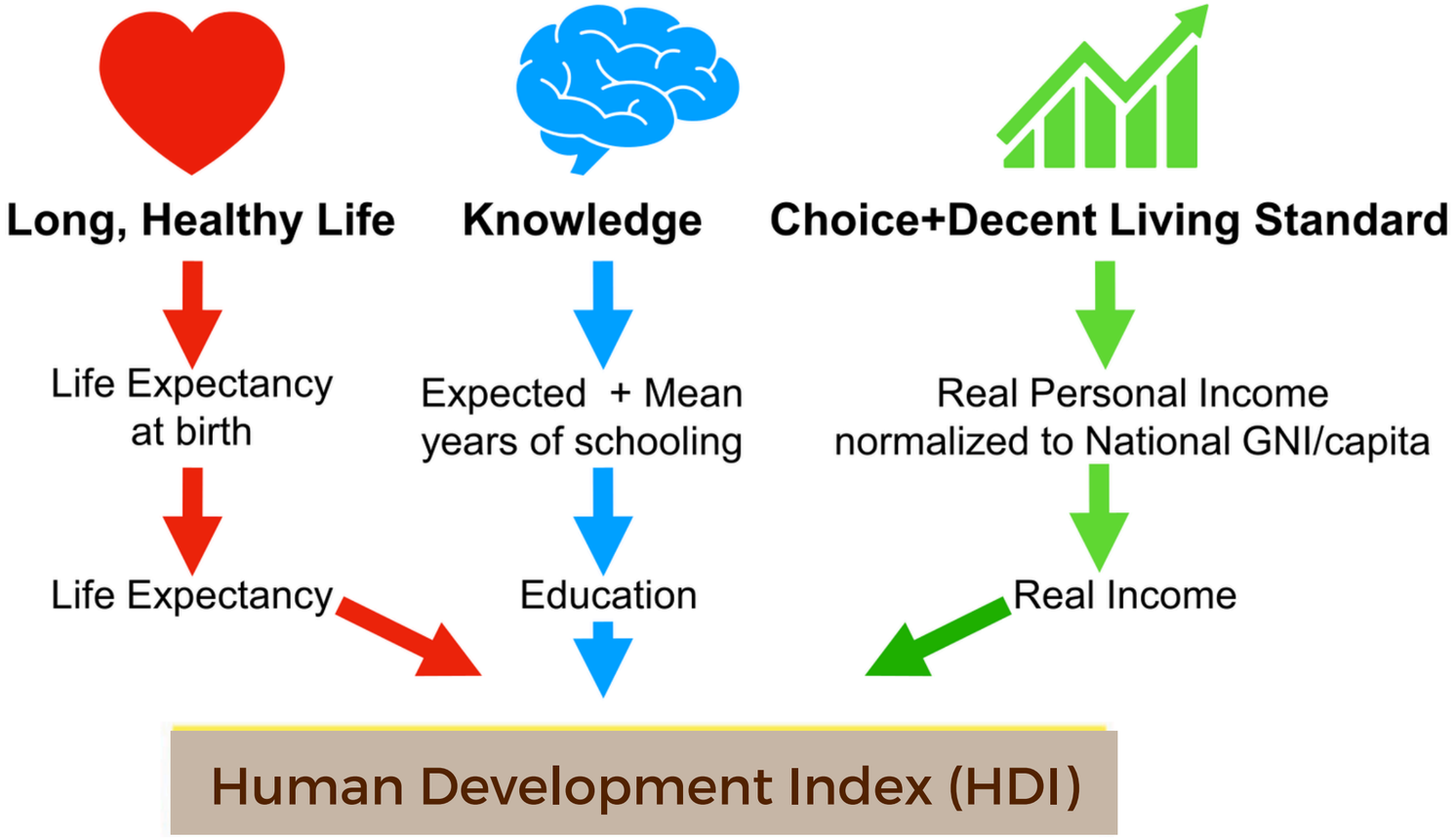
First impression,
it’s a simple question.

But each country
has its own unique way of
calculating, extracting, or handling
data from its administrative units.

"Therefore, it's not always easy
to match one system with another."

Regarding the **HDI**

In general, we TRY **adhere** to the **standard definition** by the United Nations Development Programme (UNDP)

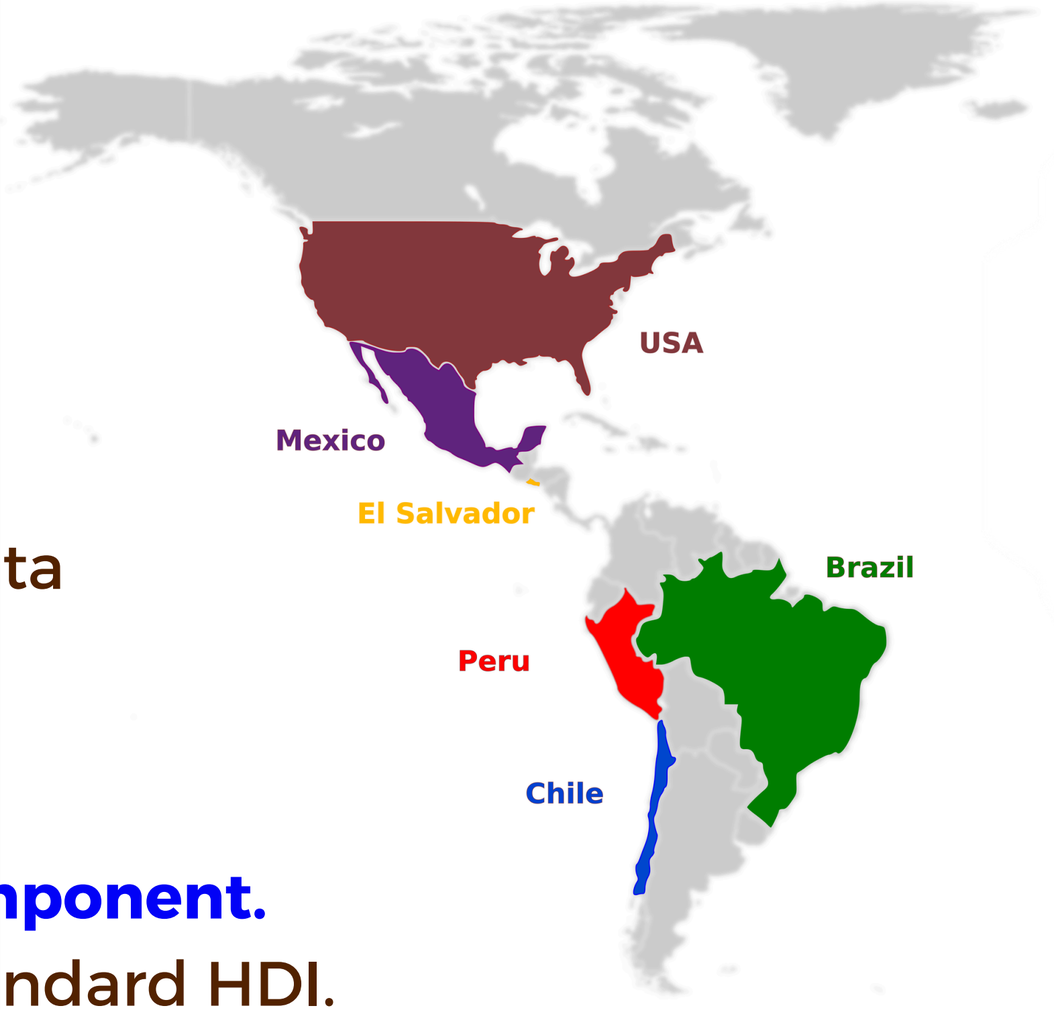


Challenges:

- **USA use “Tracts” and not cities,**
  for extracting and handling your dataset.

⬇

Solution: we convert data tracts in data cities, with Data Engineering.



- **Chile reduce** the **values** of **Economic component.**
  Generating final values lower than the standard HDI.

⬇

Solution?? Very difficult to find a certain one;

"So, what is our attempt at a solution?"

--> In short:

- compare different techniques.

--> To elaborate:

1. Search and detect Scale-Free Patterns;
2. **Create different forms of representation for comparison;**
3. Determine numerical and visual parameters to analyze the results.

Methodology for ScatterPlots:

Government data table | published research papers

Dataframe:

```
def botao_menu_dos_dataframes():
    button = widgets.ToggleButton(value=False, description='Click me', disabled=False,
                                  button_style='success', # 'success', 'info', 'warning', 'danger' or ''
                                  tooltip='Description', icon='check')

    out = widgets.interactive_output(click, {'button': button})
    display(widgets.VBox([widgets.HBox([button]), out]))

# Chama Menu Interativo dos Dataframes (com botão)
botao_menu_dos_dataframes()
```

✔ Click me

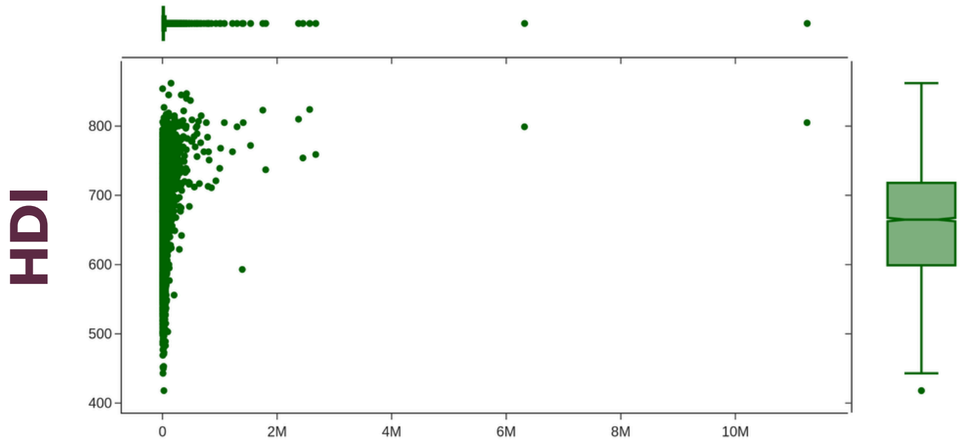
Dataframe df_Censo2010

Min_Linha 7 Max_Linha 143

	Municipio	UF	Macroregioes	IDHM_2010	IDHM_Renda_2010	IDHM_Longevidade_2010	IDHM_Educação_2010	Total_Pop_Absol	Total_Pop_Urbana_Ab
7	Abatiá PR	PR	Sul	687	676	804	596	7764	5
8	Abaíra BA	BA	Nordeste	603	577	746	510	8316	3
9	Abdon	SC	Sul	694	660	812	625	2653	

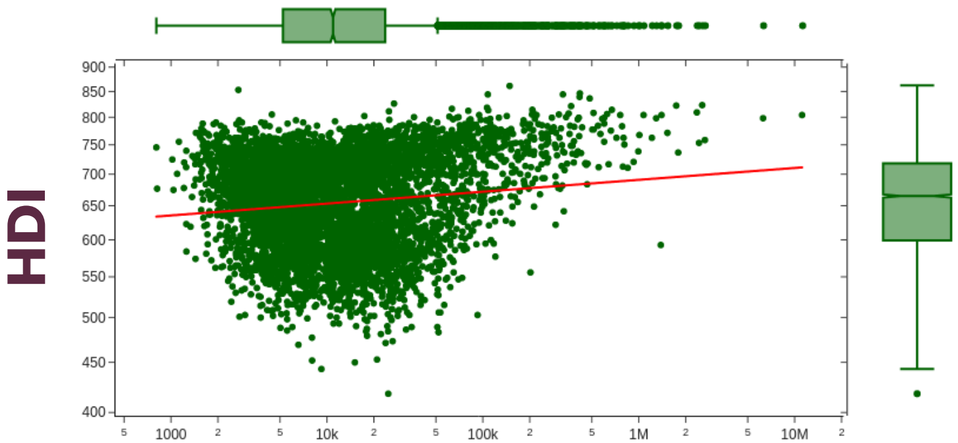
Case: **Brazil** Normal Scale

(HDI x Population)

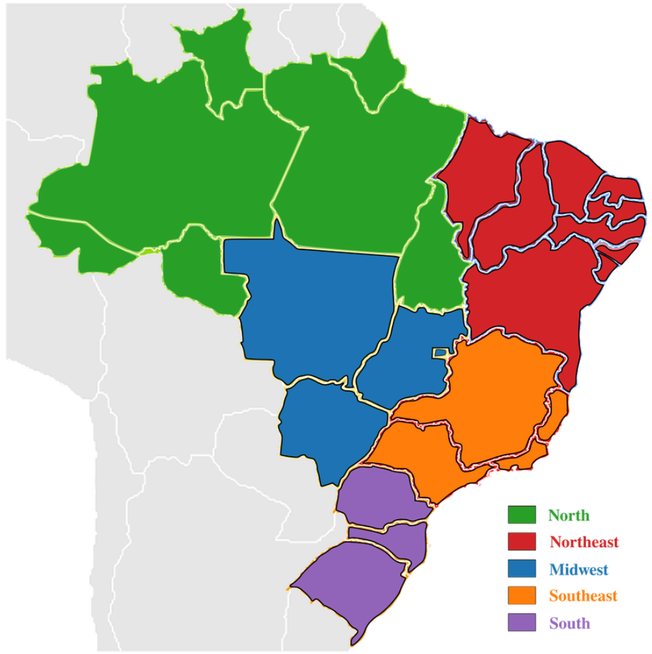


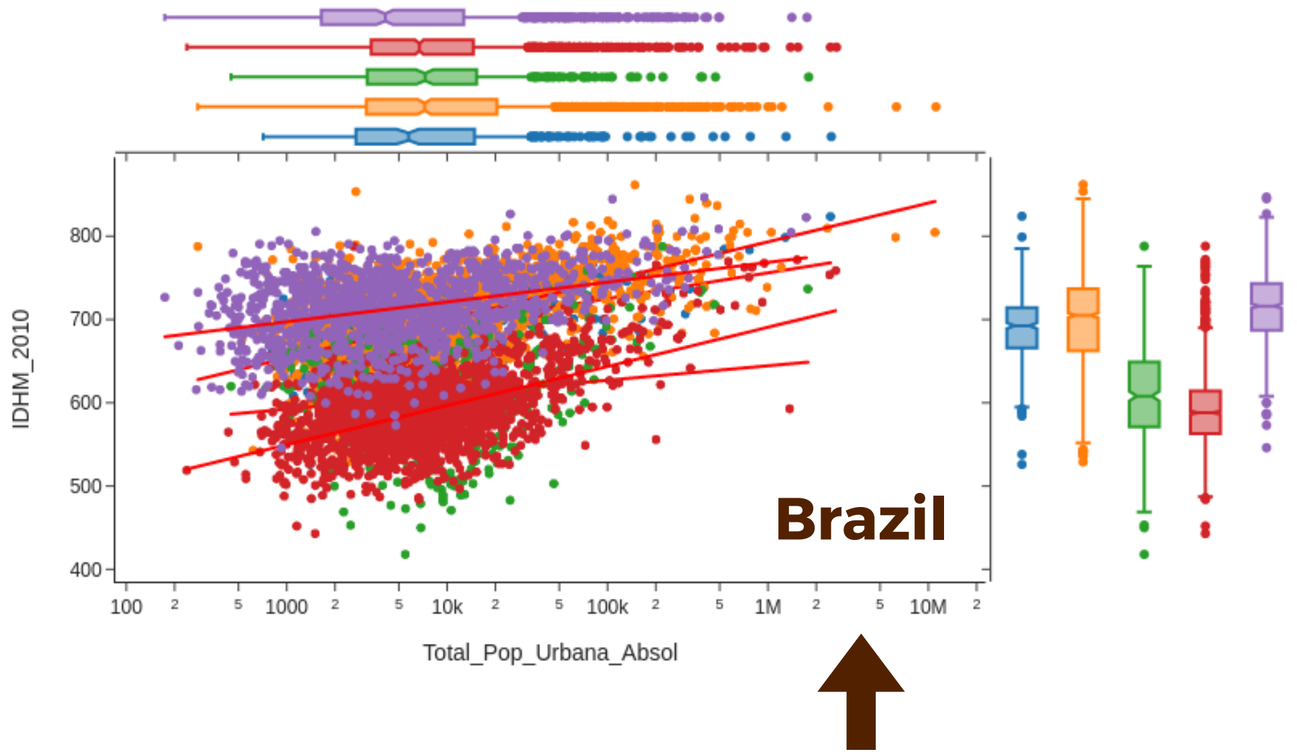
Brazil Log-Log Scale

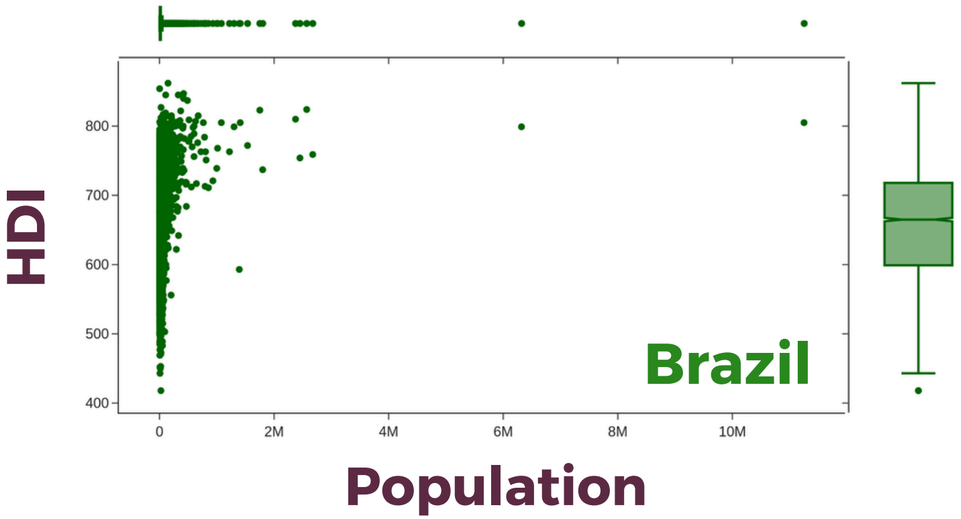
(HDI x Population)

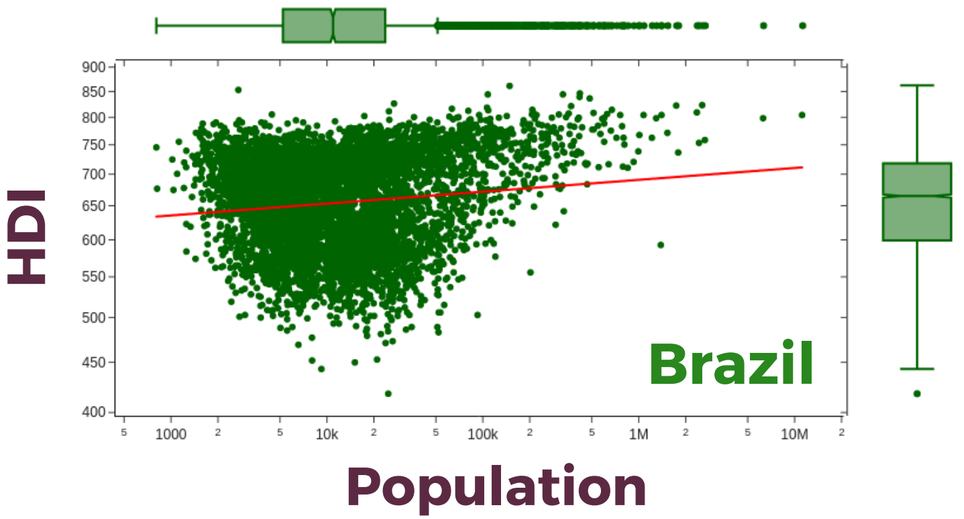


National Scale

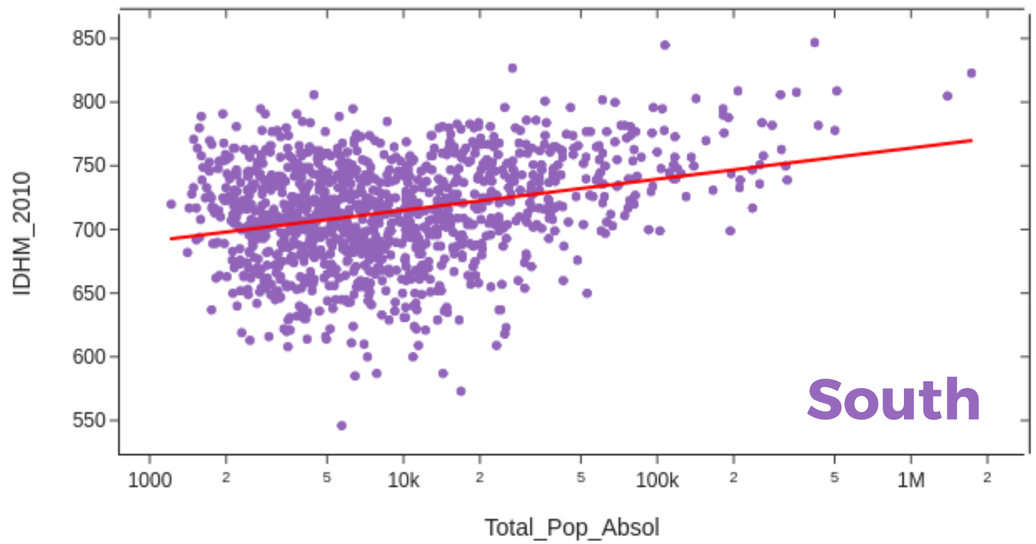


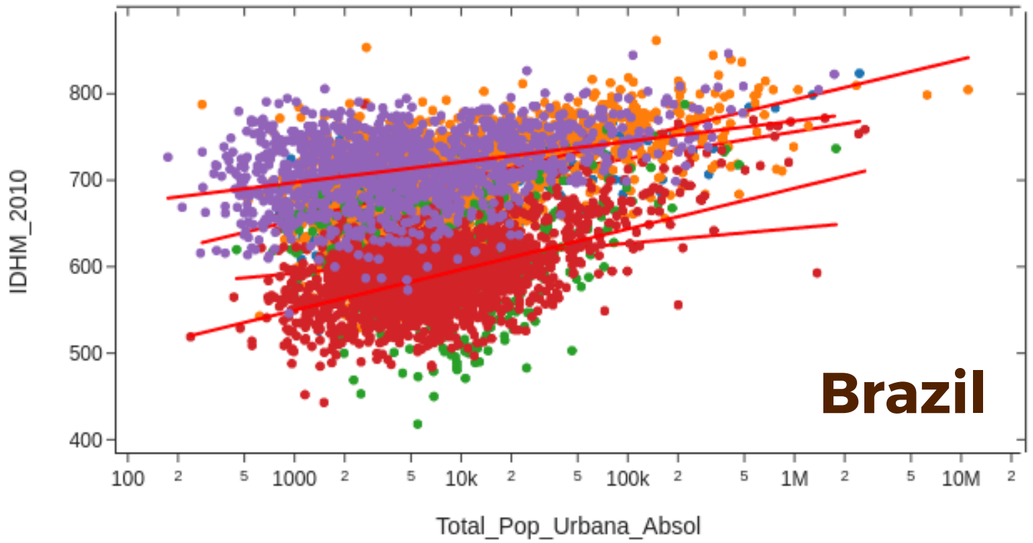


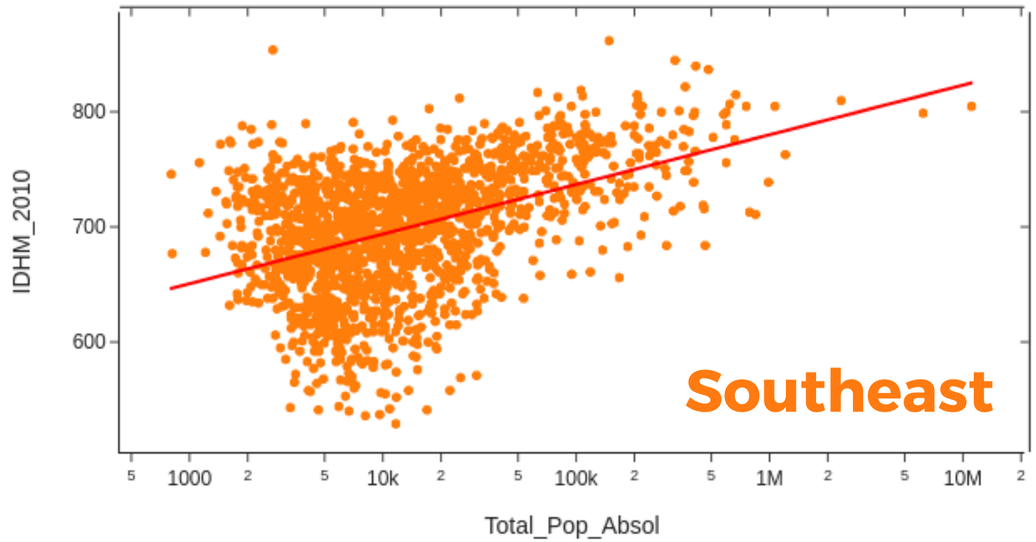


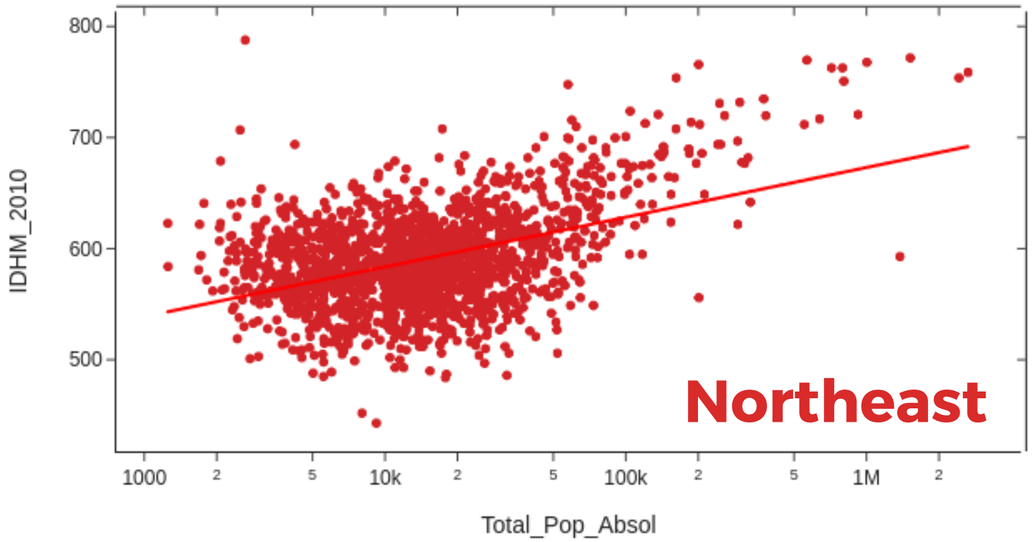


Nacional Scale -> MacroRegional Scale

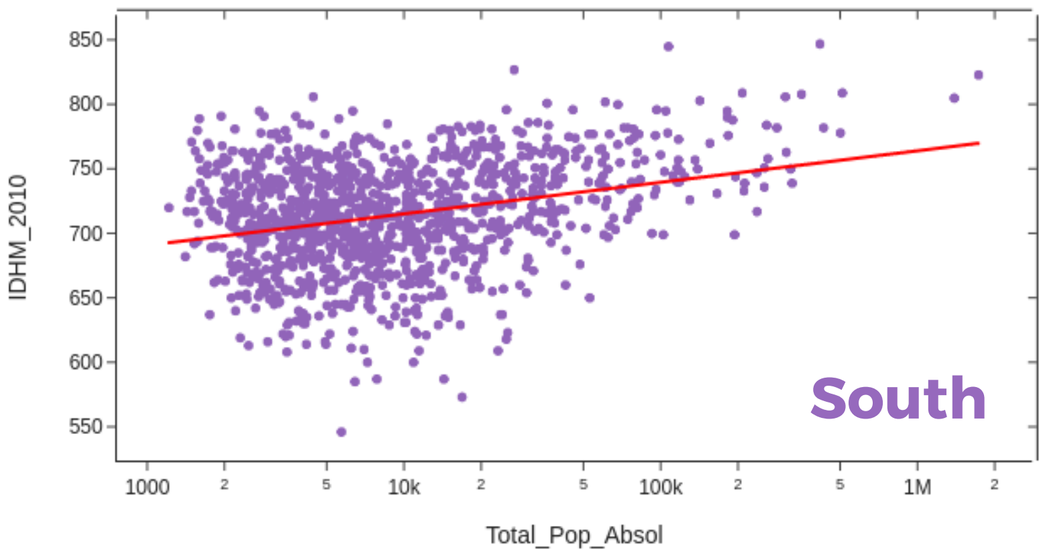


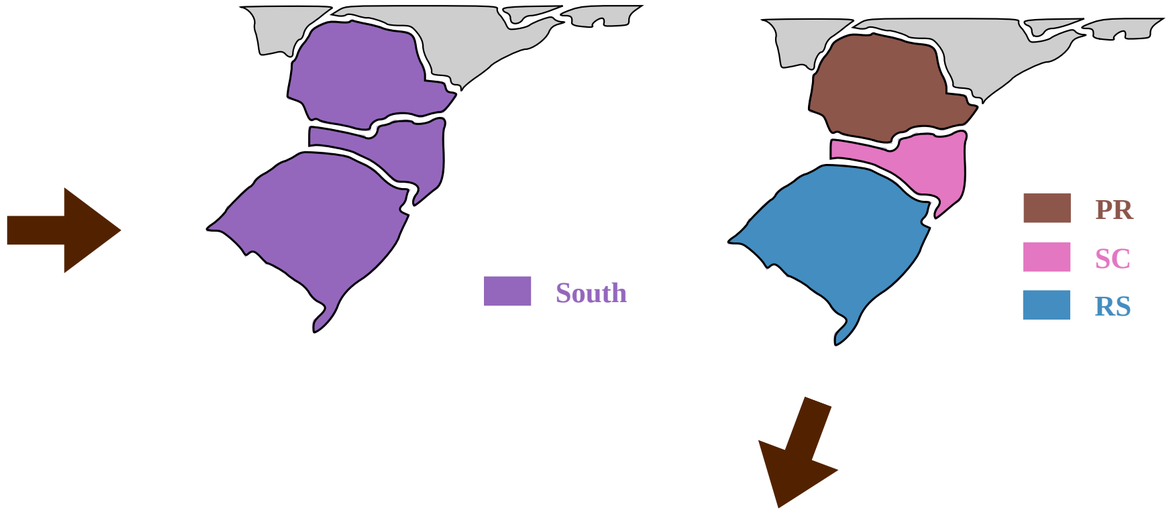


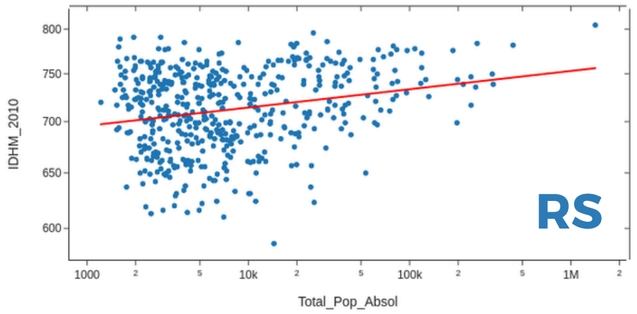


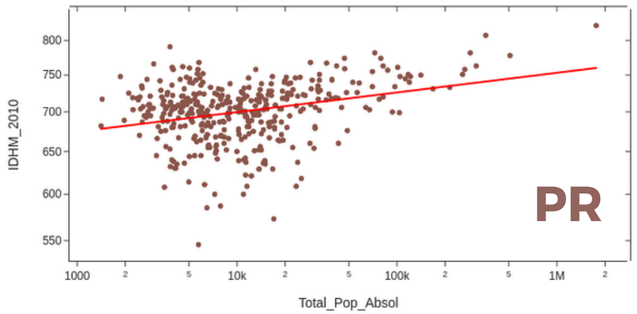


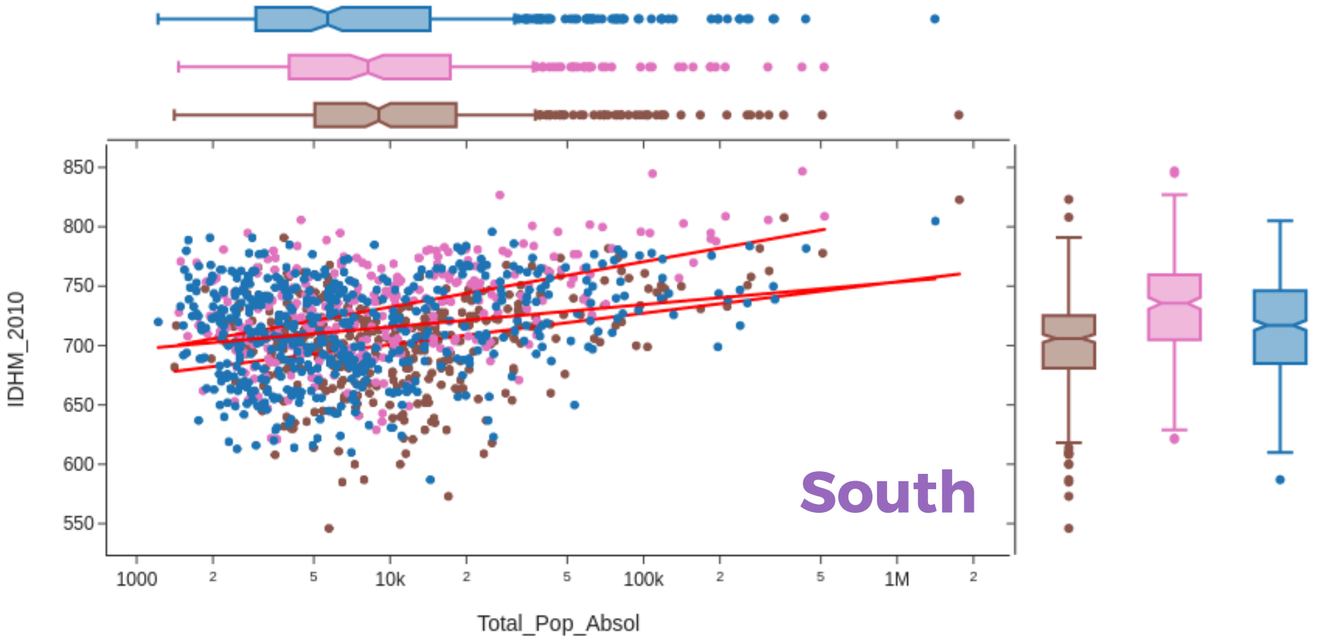
MacroRegional Scale -> Regional Scale







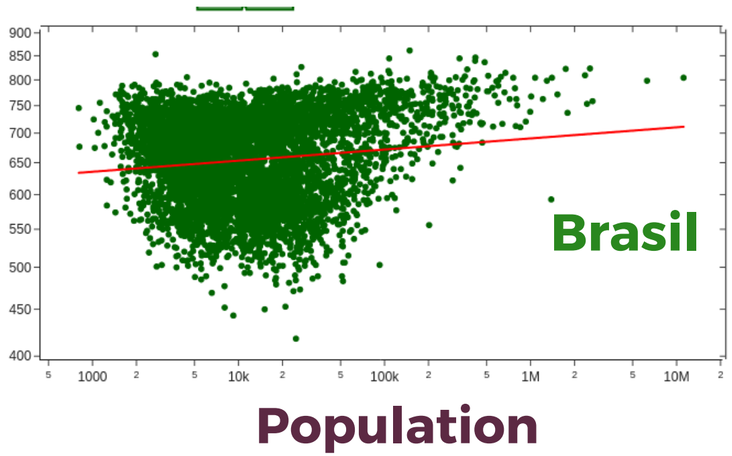


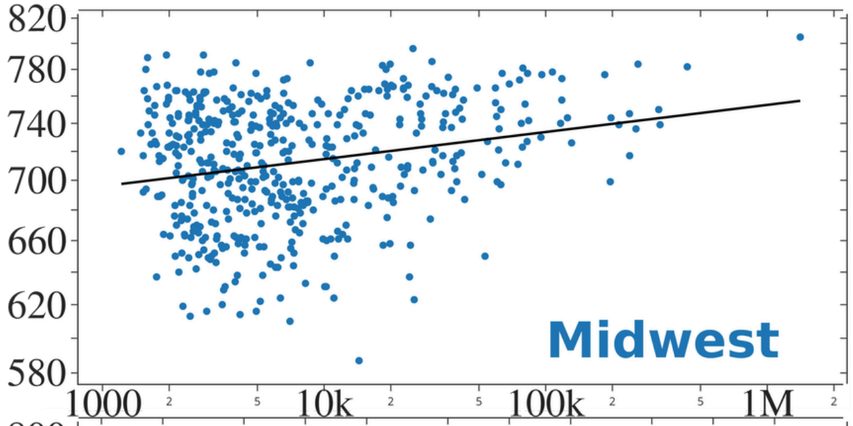


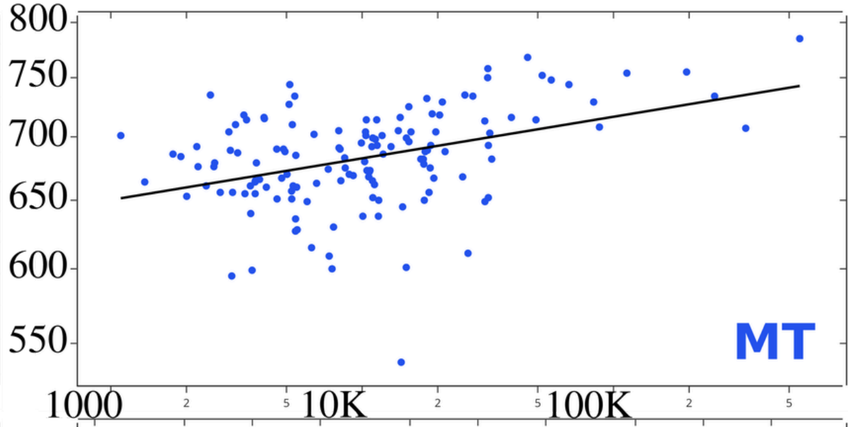
Nacional Scale

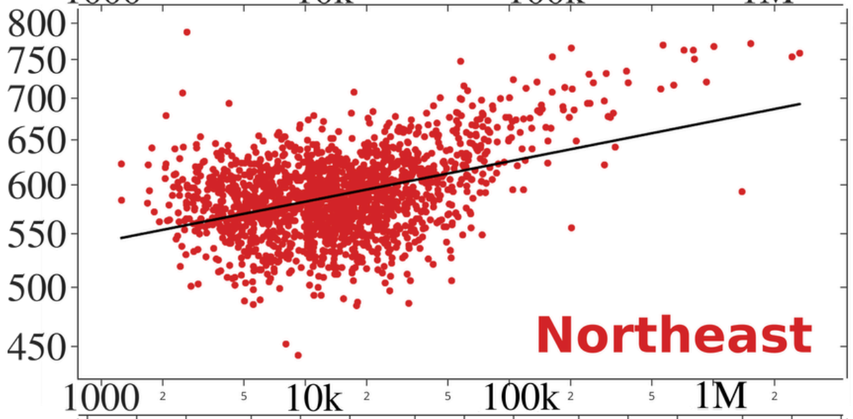
Macroregional Scale

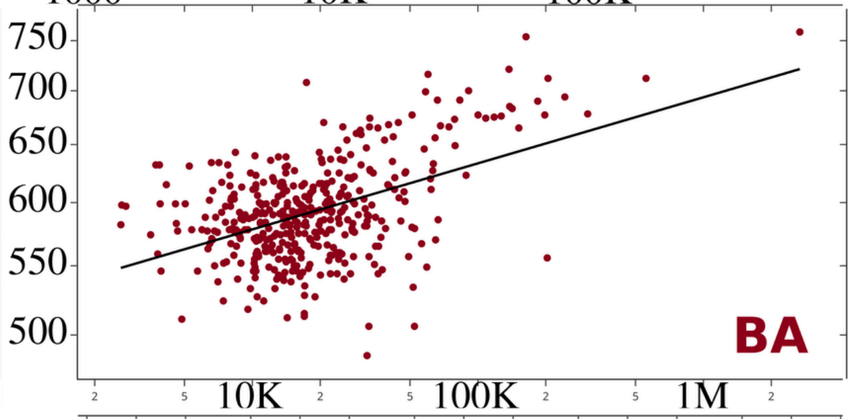
Regional Scale

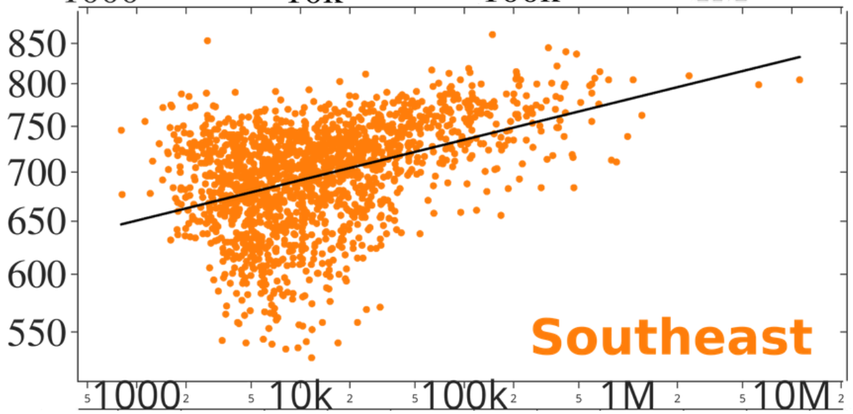


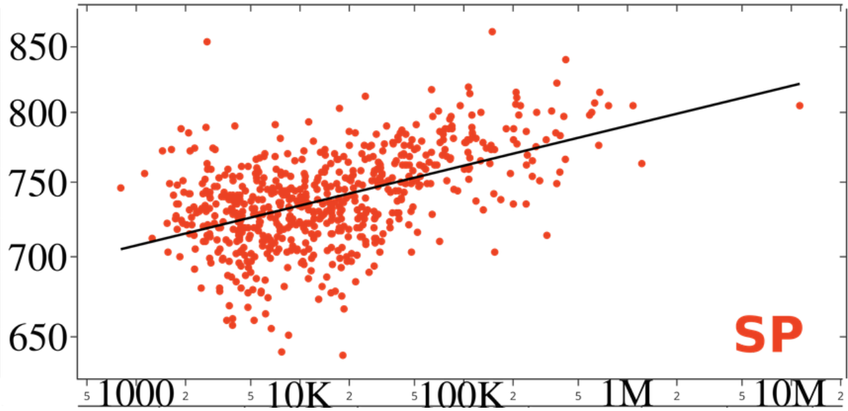










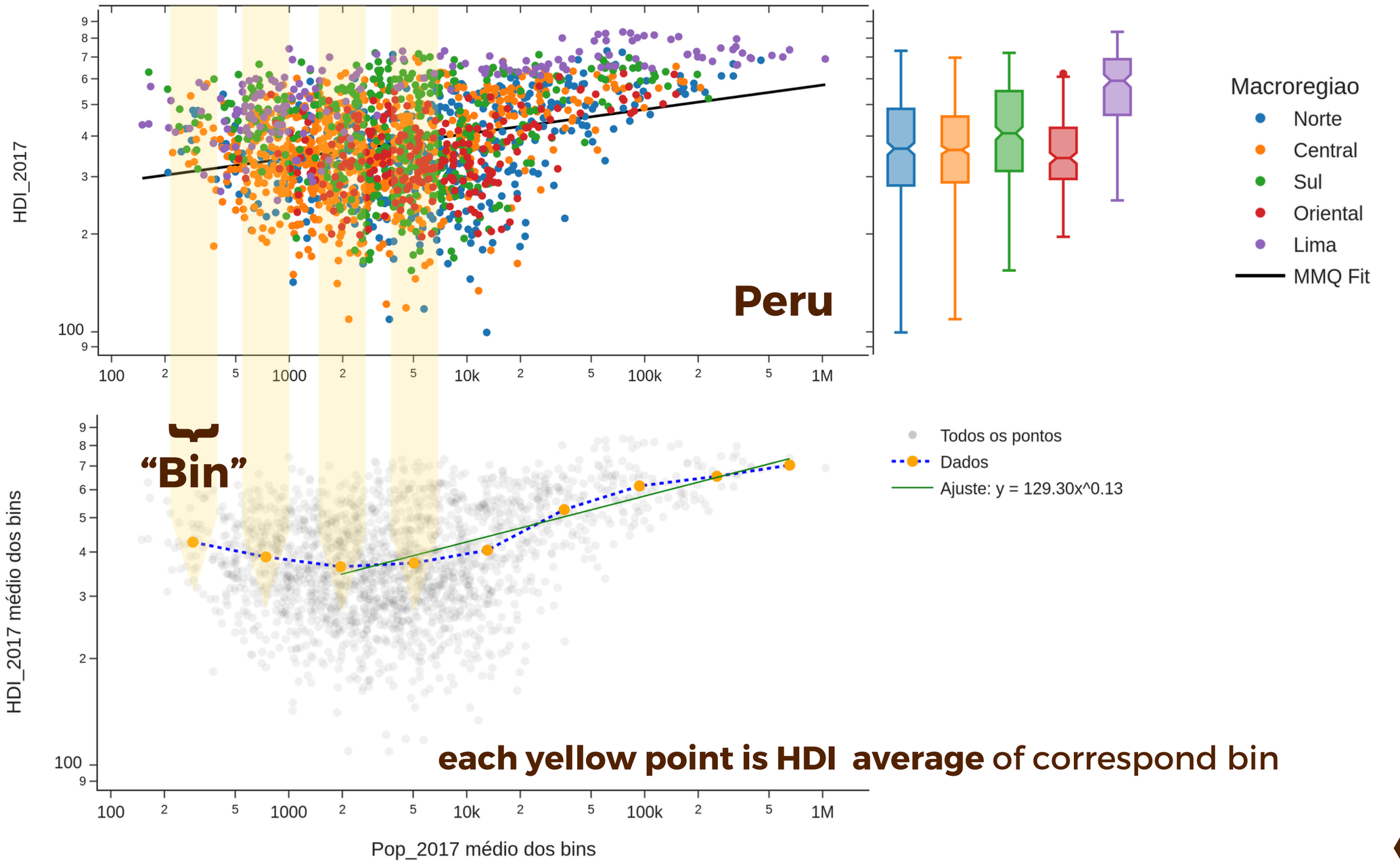


How to reduce noise from these Scatter Plots?

--> One way would be: **Binscatter Process**

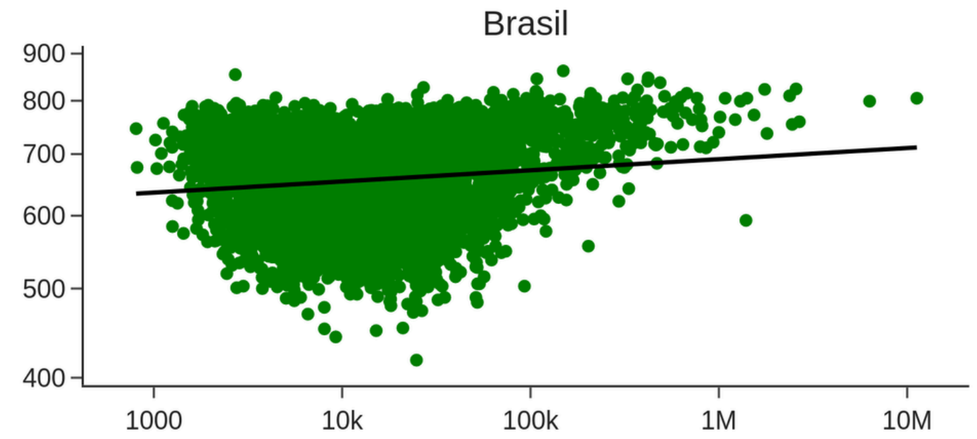
Case: Peru

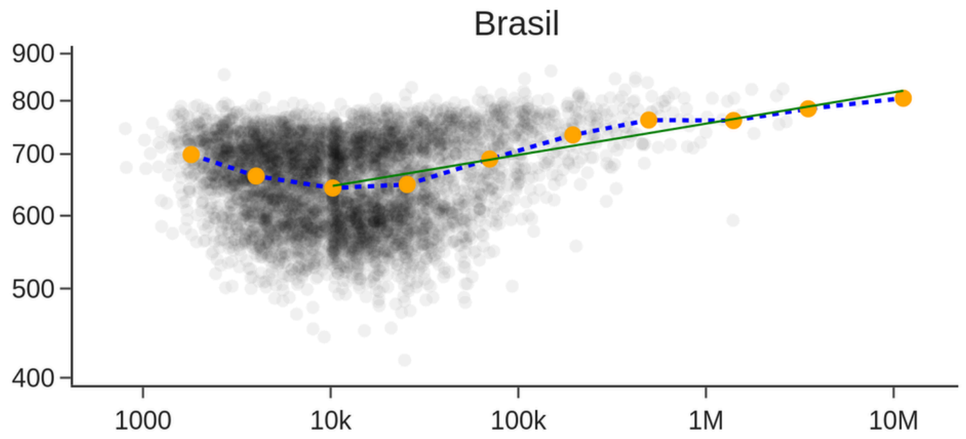
Divided by “Bins”



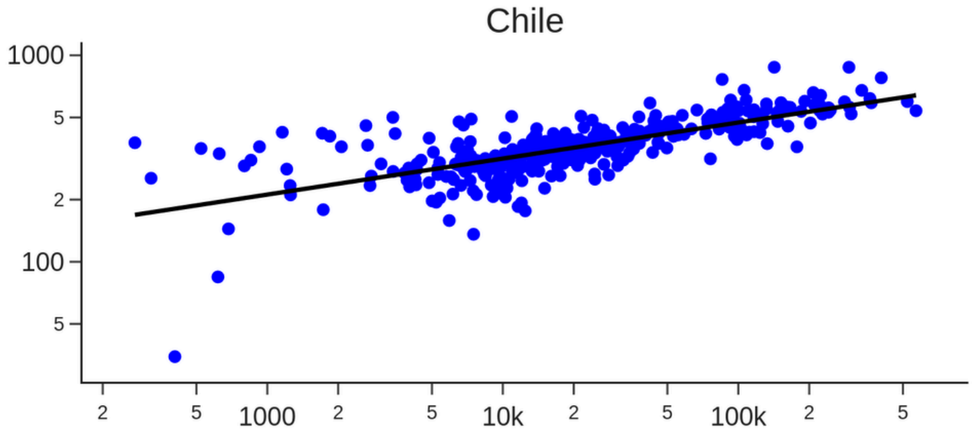
each yellow point is HDI average of correspond bin

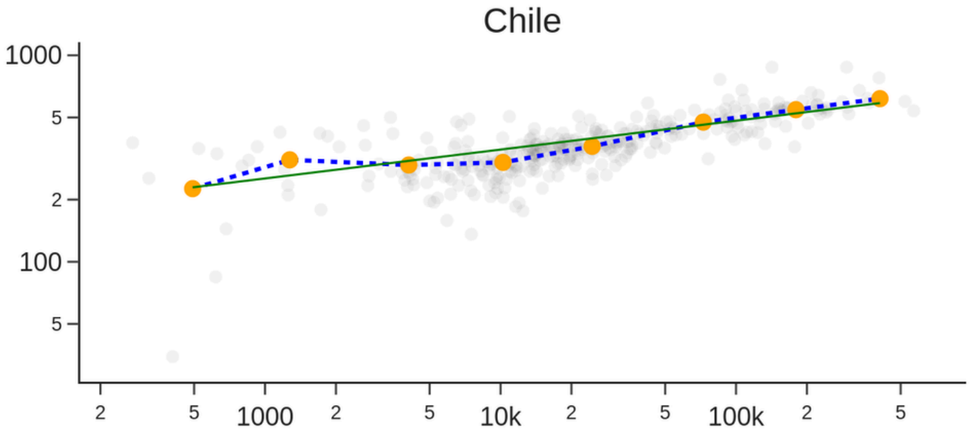
Brazil



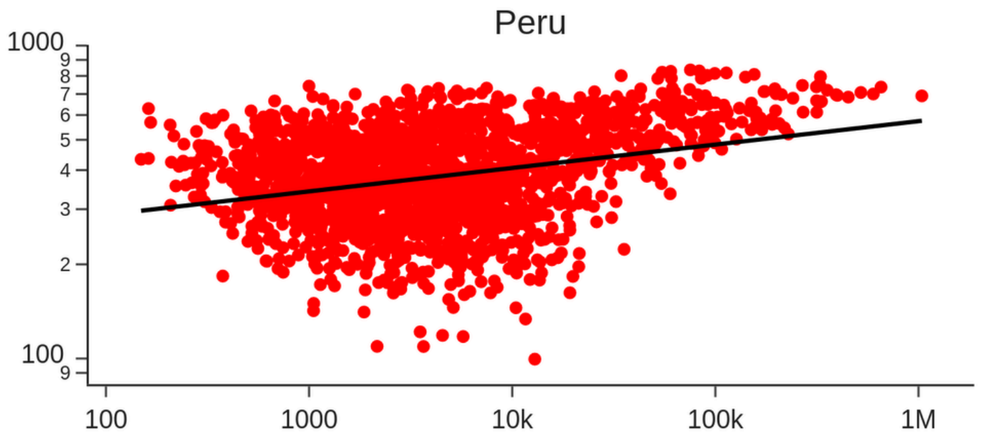


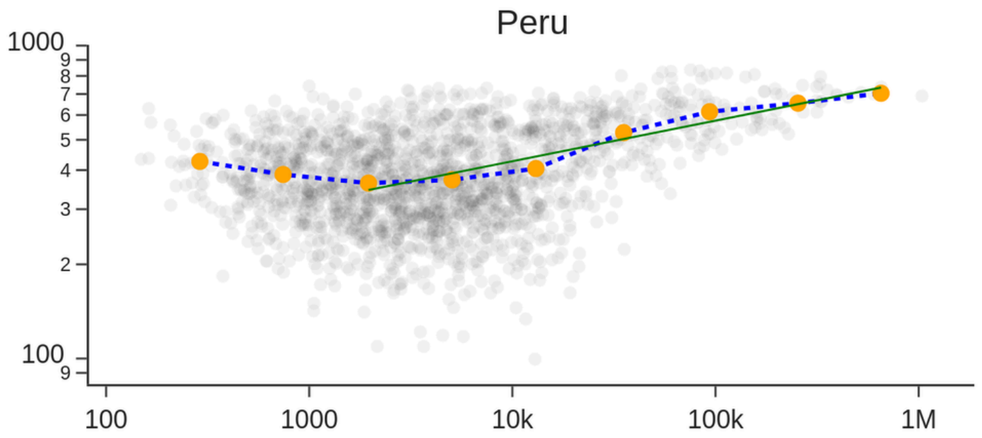
Chile



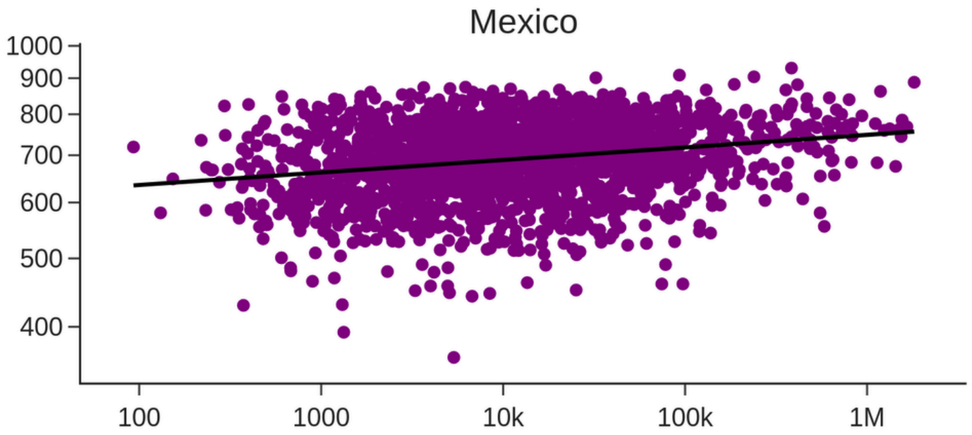


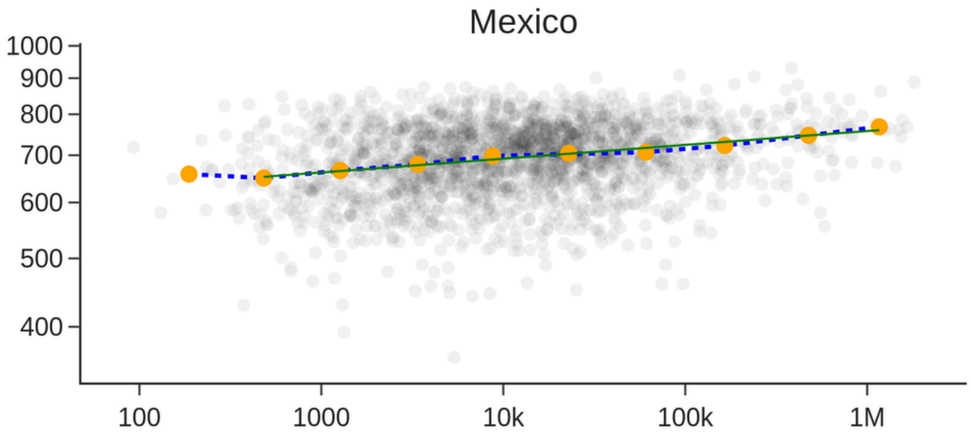
Peru



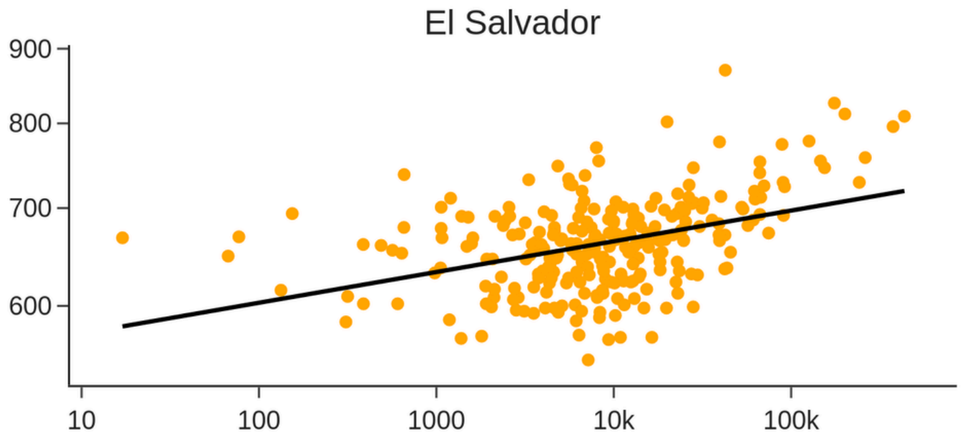


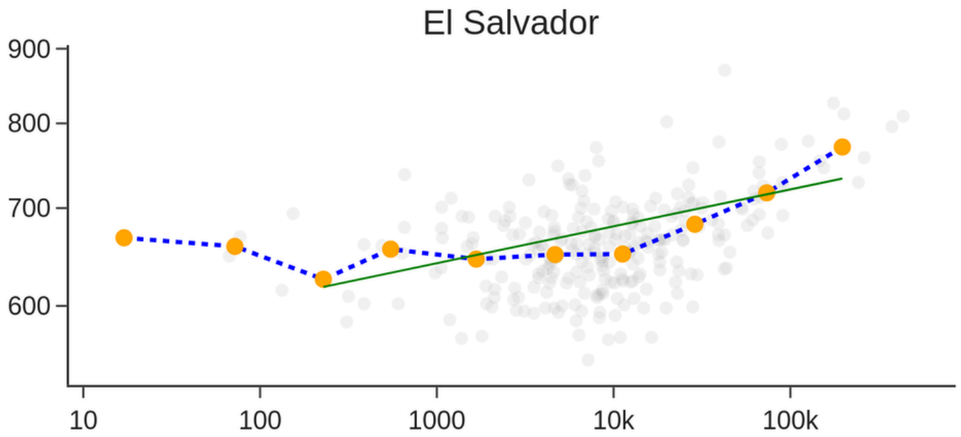
Mexico



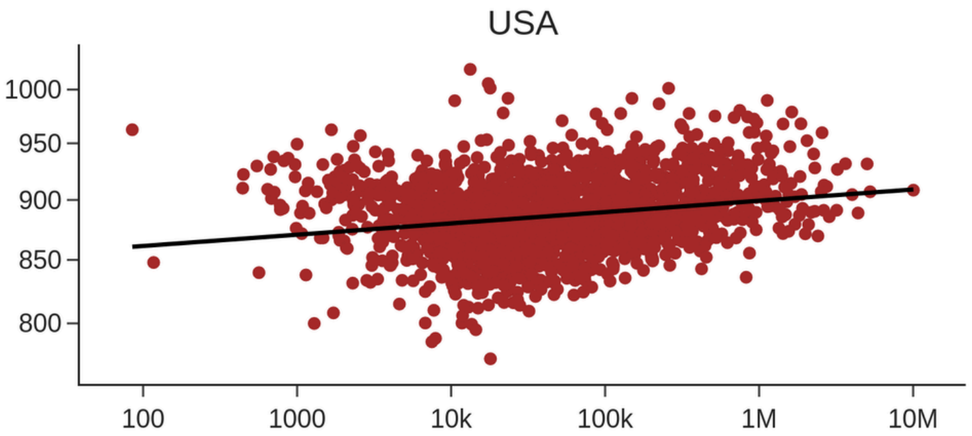


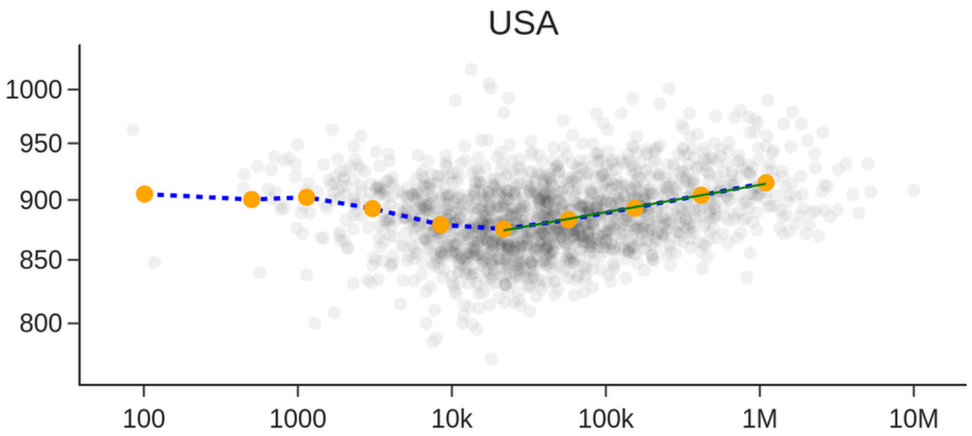
El Salvador





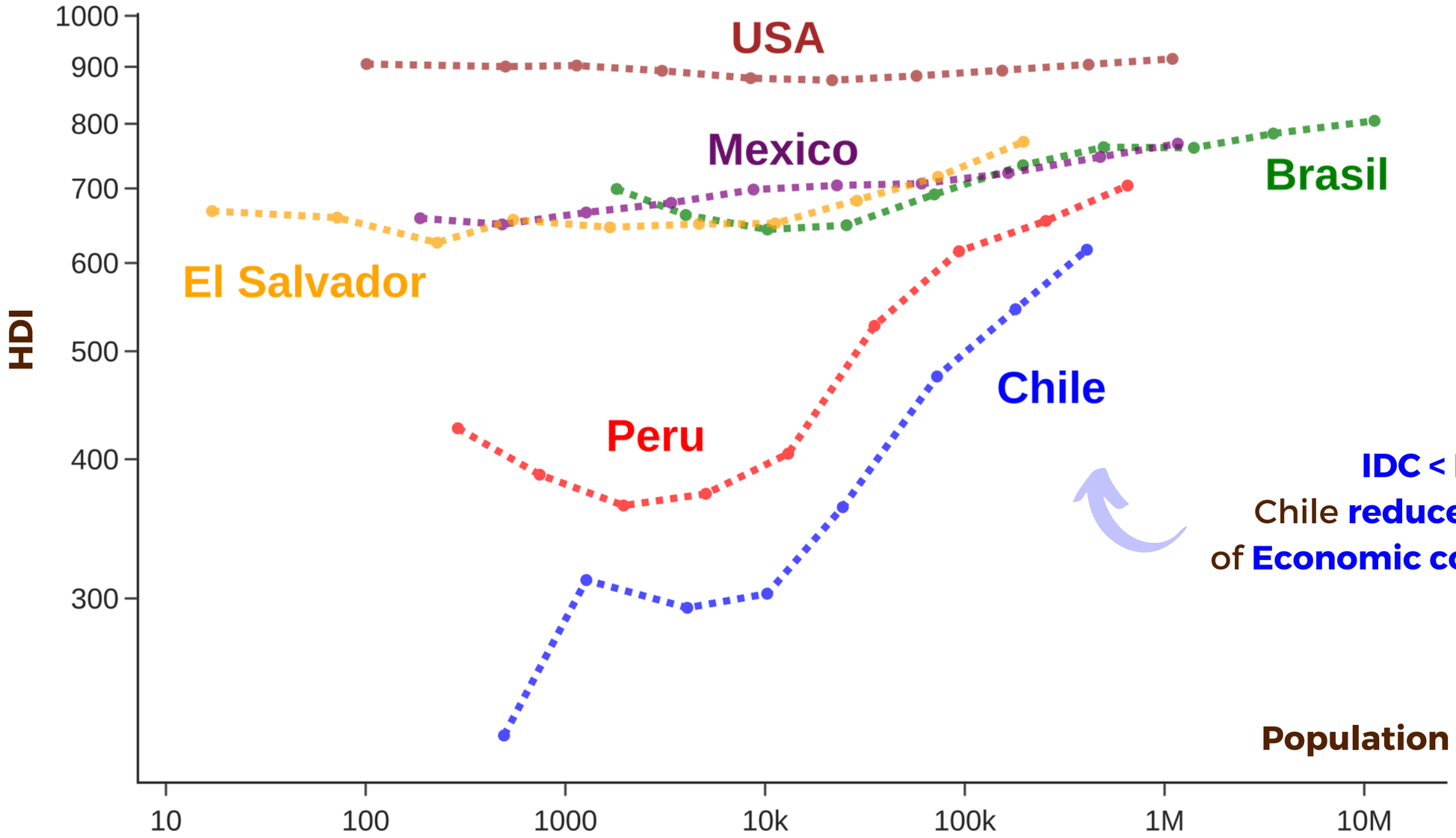
USA





Some compared results

"The **bin division** is **adjusted** based on the population variation of **each country**."

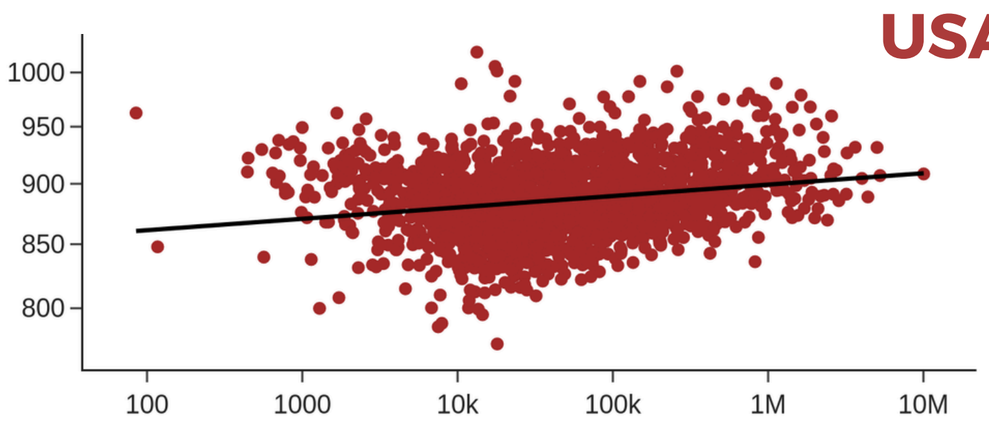


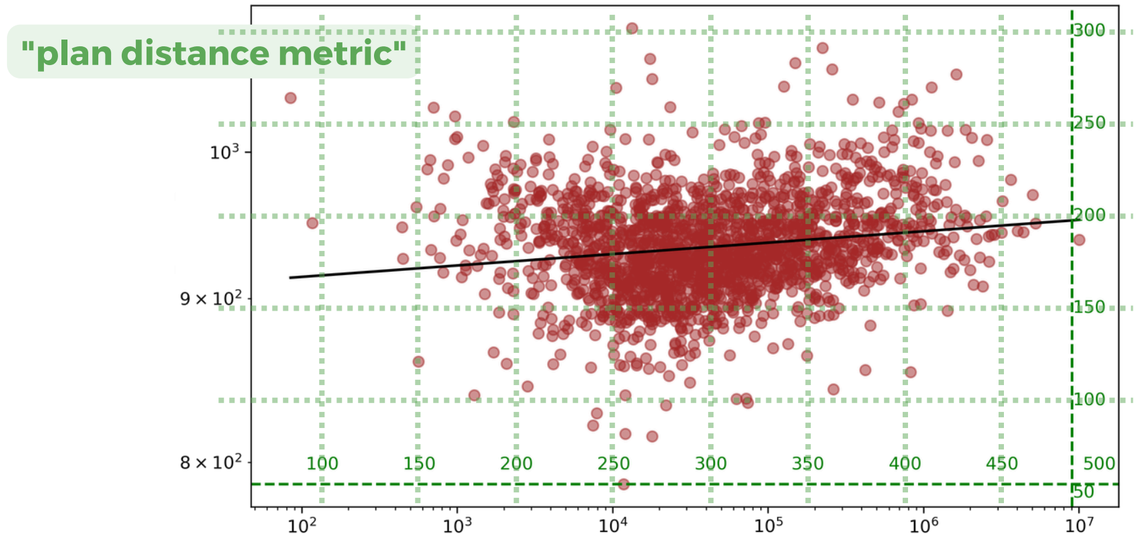
IDC < HDI
Chile **reduce values** of **Economic component.**

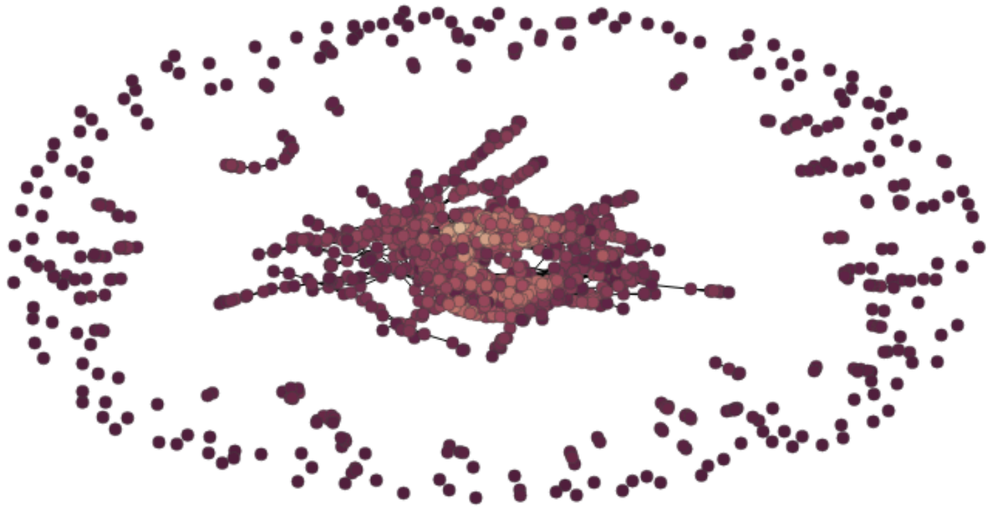
How to represent this Scatter Plots in Network graph?

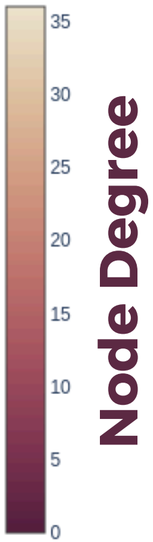
Based on GCA: Gravitional Cluster Algorithm

"Pixel Axes" : 500 x 300 px

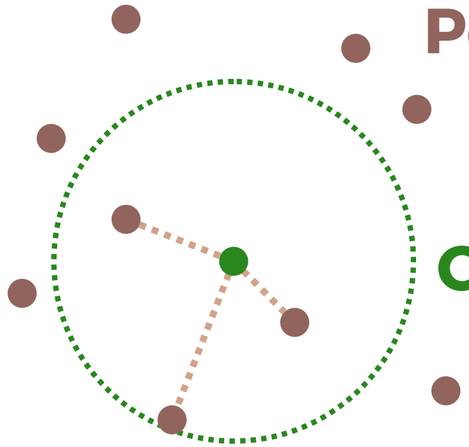








Cut Radius = 5



Point = City

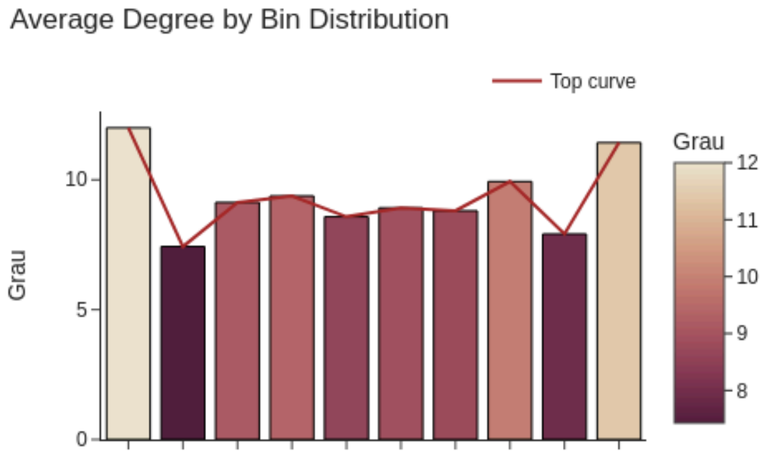
Cut Radius

Edge = Distance on map

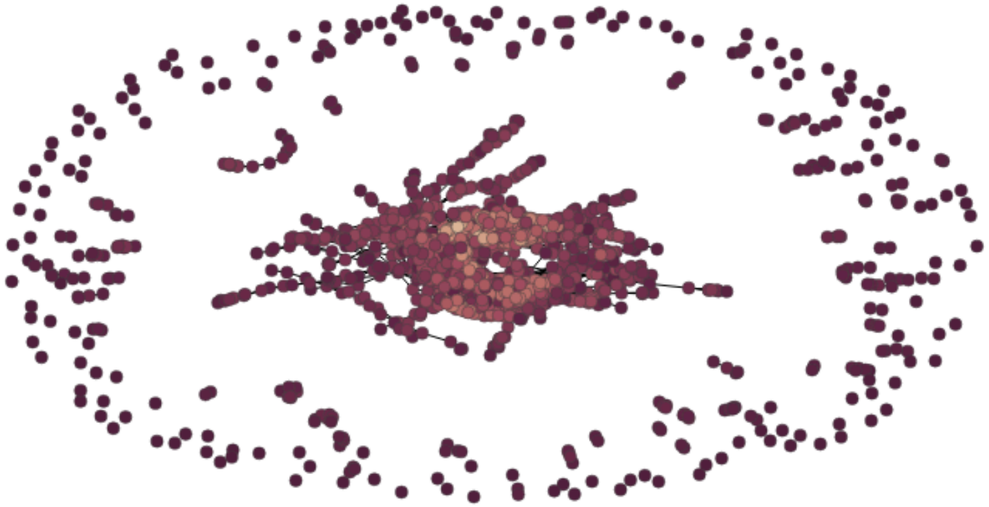
Based on GCA: Gravitional Cluster Algorithm

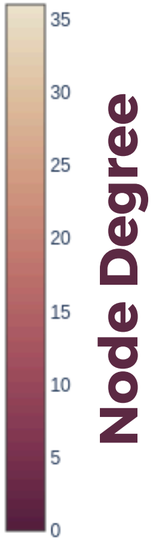
we may explore

statistical information

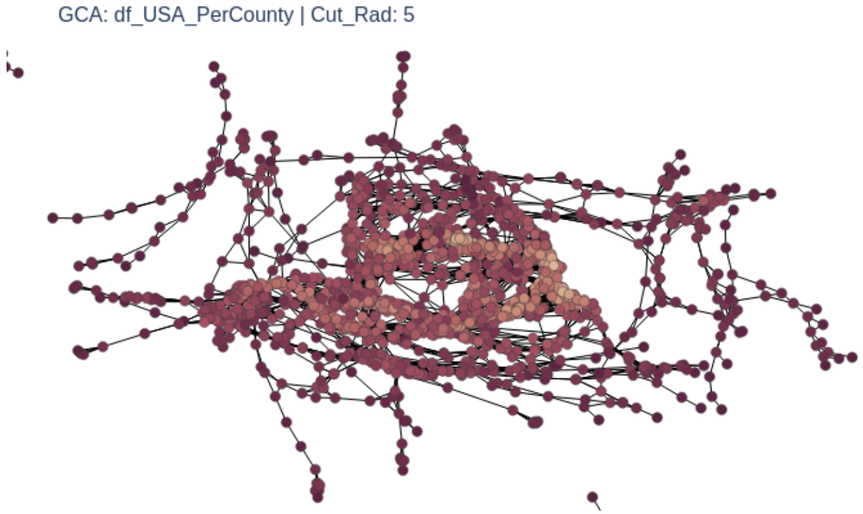


Metric	Value
Nº of nodes	1872.0
Nº of edges	8359.0
Average degree	8.9
Max. degree	36.0
Min. degree	0.0
Variance	46.8
Standard deviation	6.8



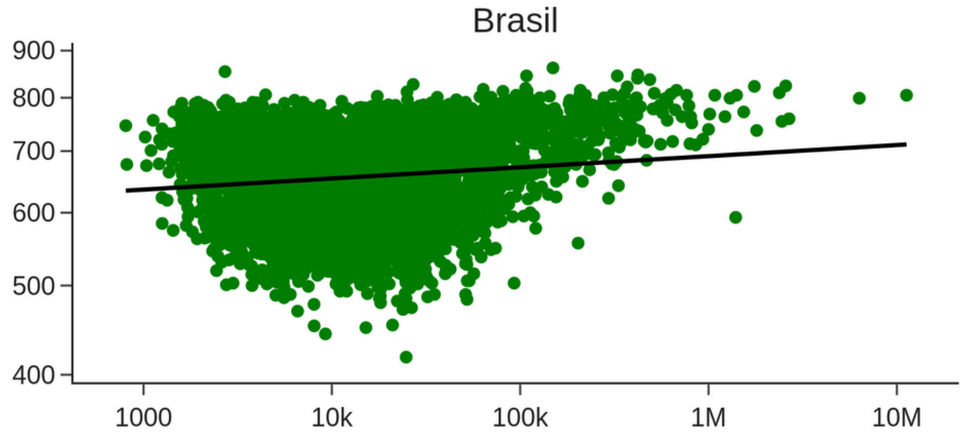


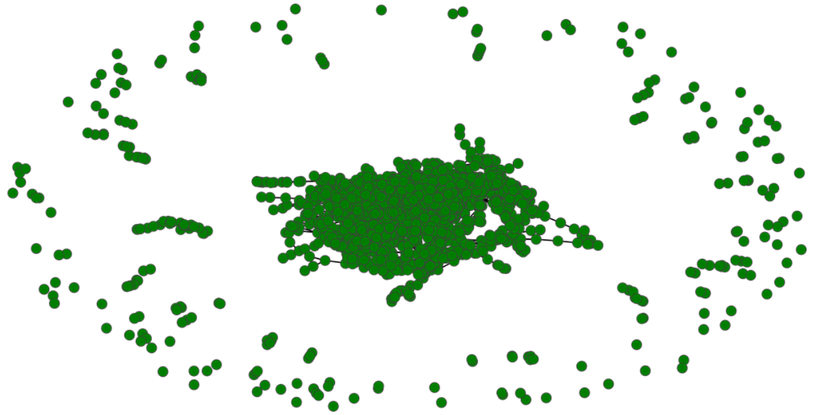
Node Degree



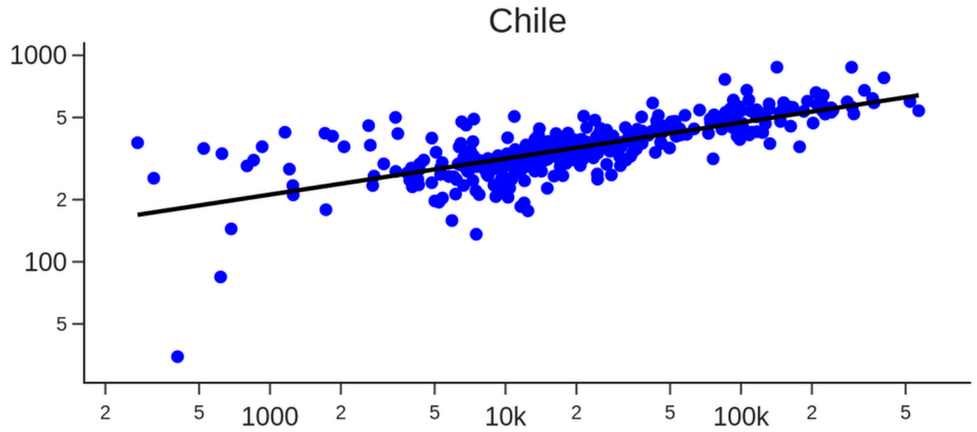
Node Degree > 0

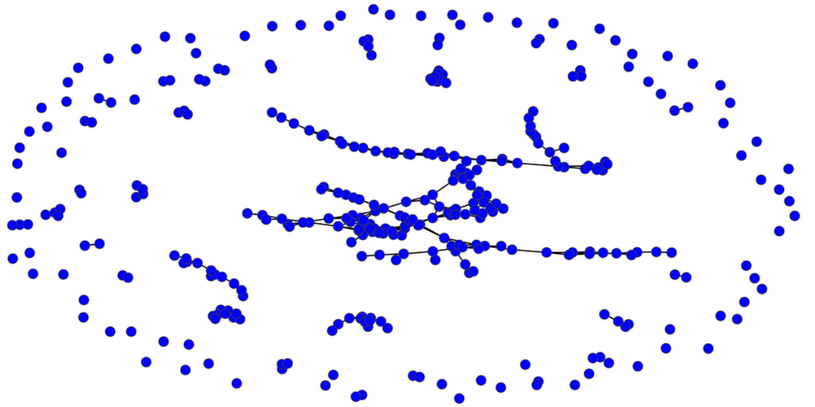
Brazil



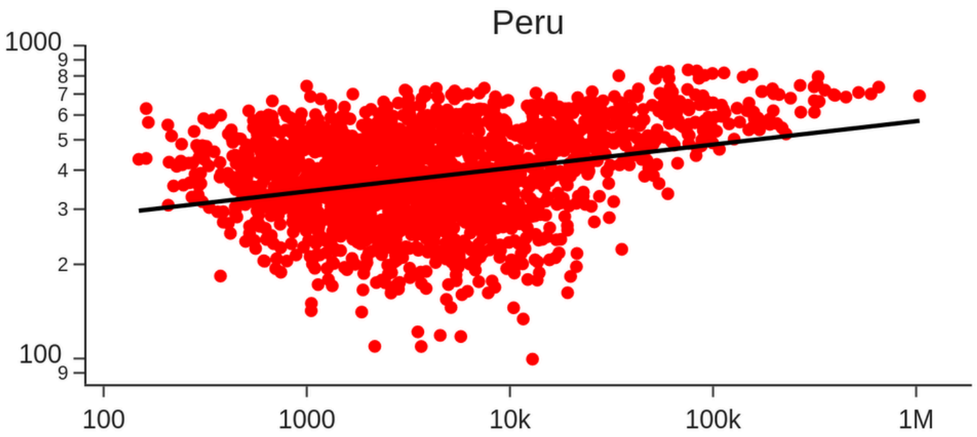


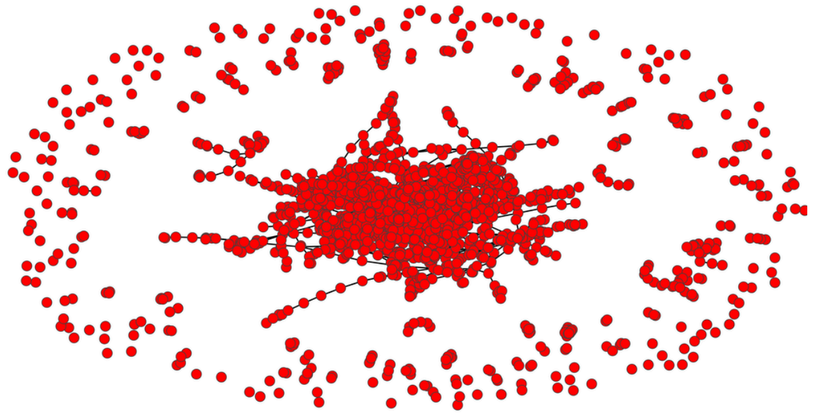
Chile



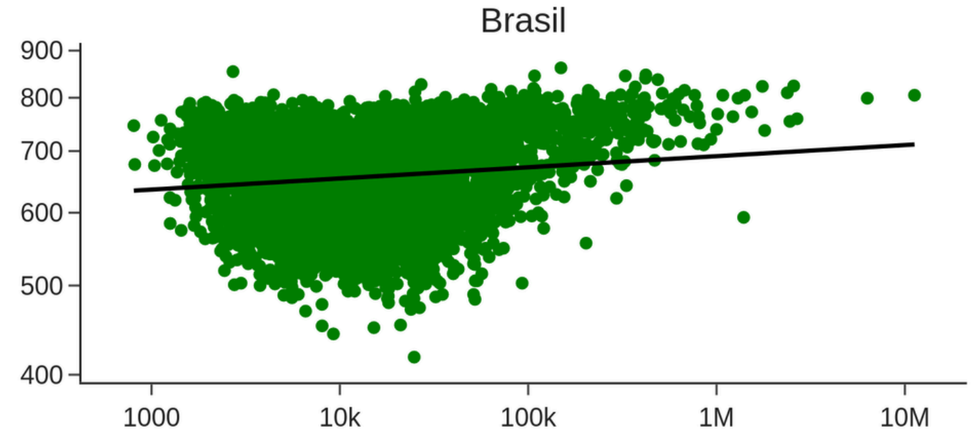


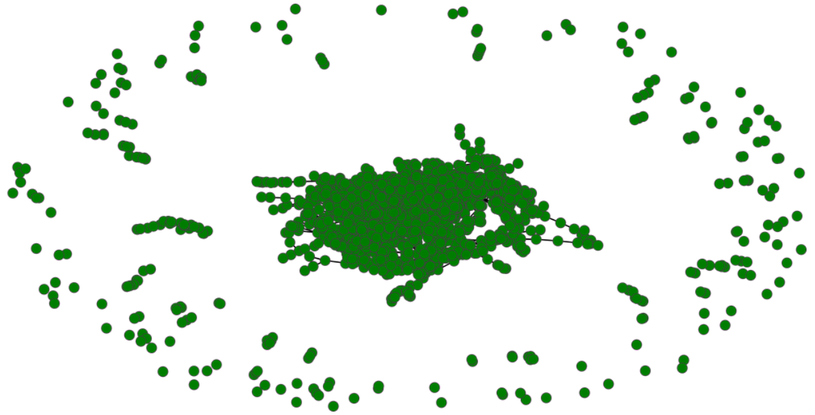
Peru





Brazil





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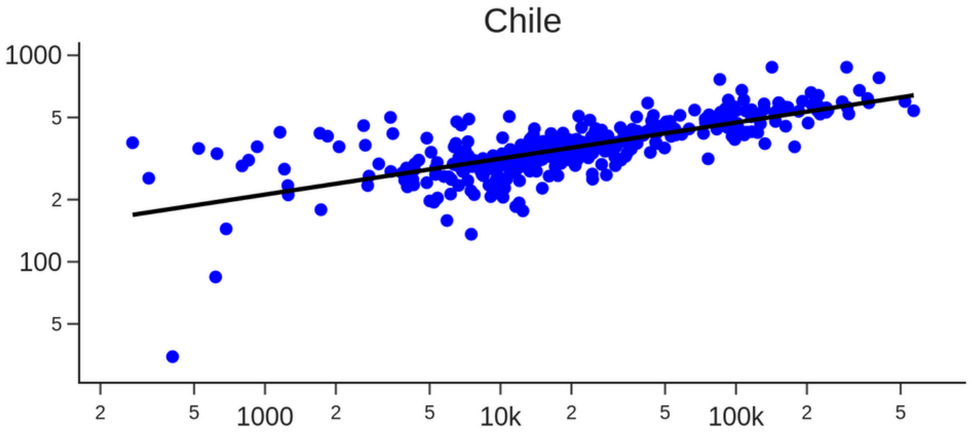
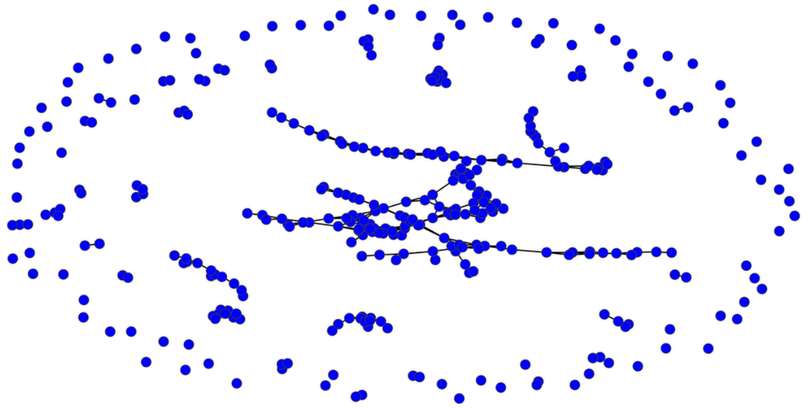
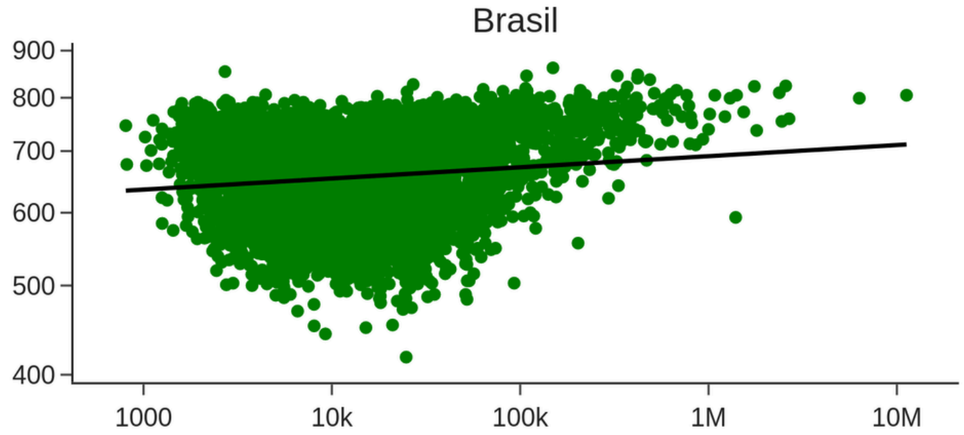
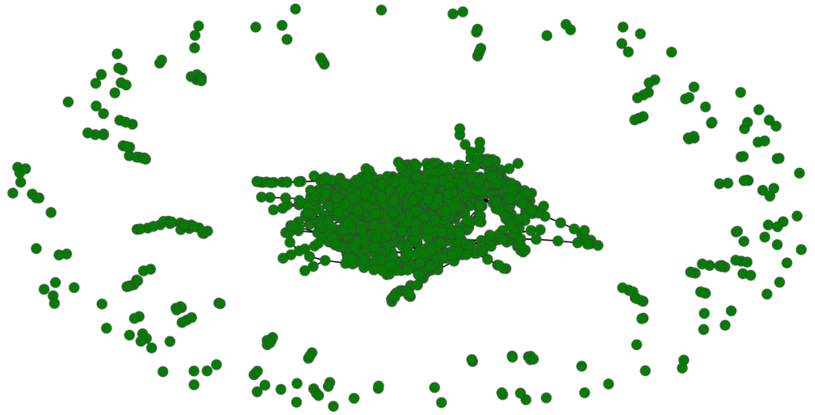




- **More** number os **points**, **more hub** in **central region** in the graph

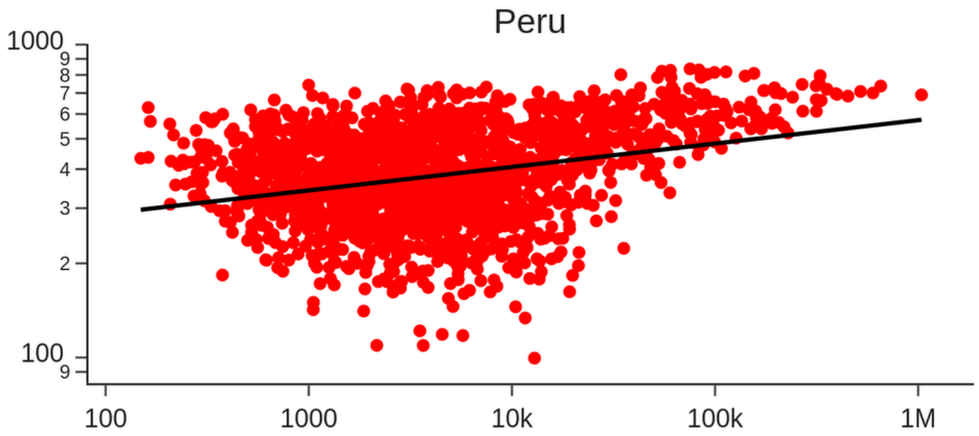
Brazil

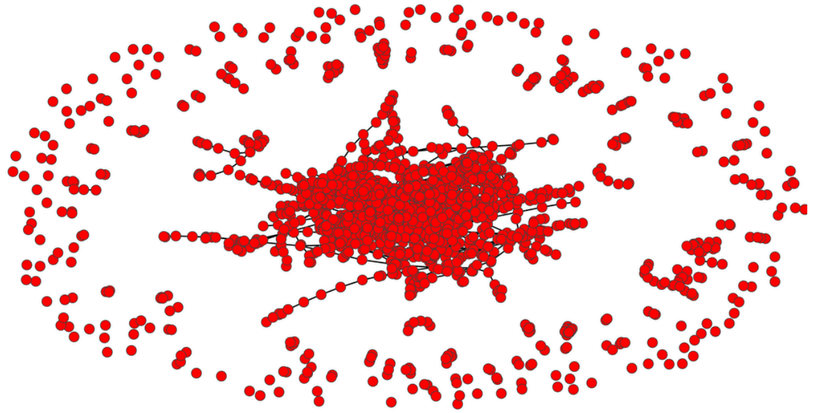




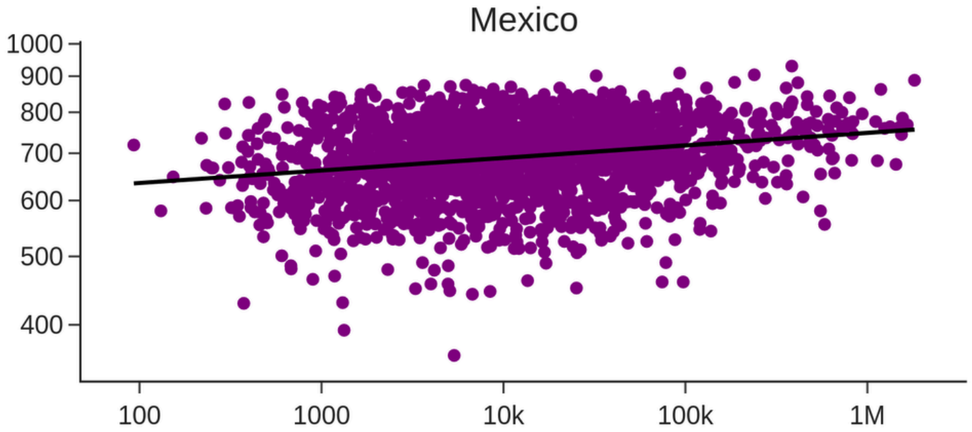
Brazil and Peru: the visual behavior is very similar, just like Mexico in relation to the USA

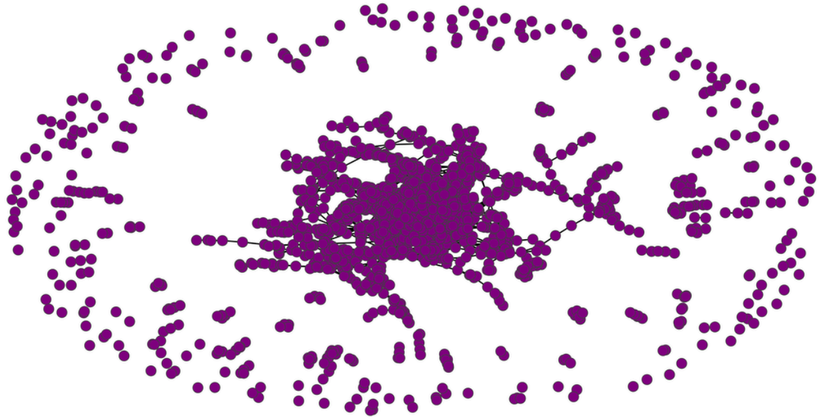
Peru



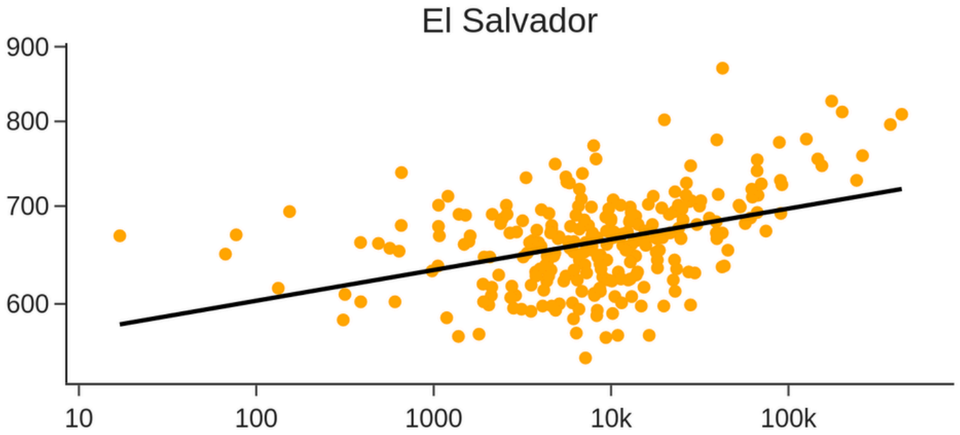


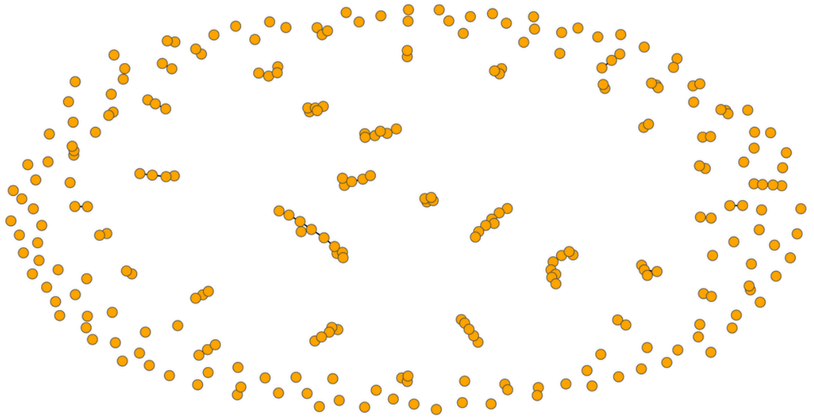
Mexico



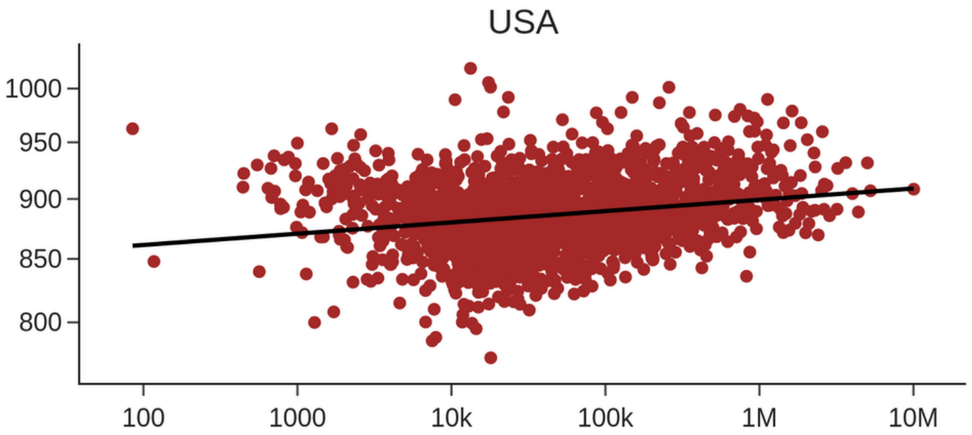


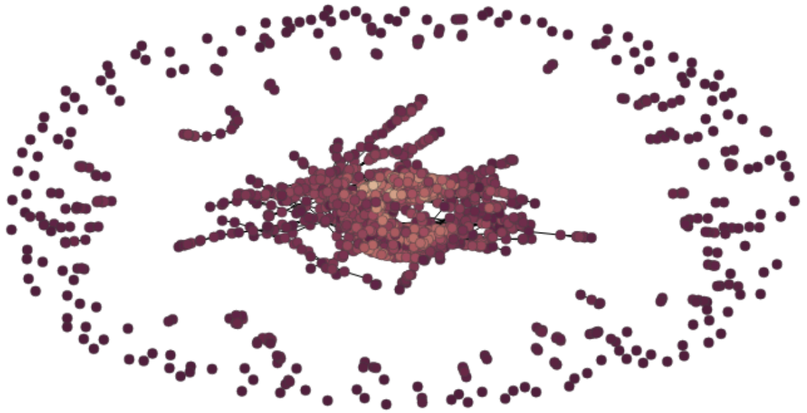
El Salvador





USA





Some Table compared results

Metric	Brazil	Chile	Peru	Mexico	ElSalvador	USA
Nº of nodes	5565	345	1874	2193	261	3142
Nº of edges	60890	456	4535	8449	112	27652
Average degree	21.9	2.6	4.8	7.7	0.9	17.6
Max. degree	59	12	15	30	4	54
Min. degree	0	0	0	0	0	0
Variance	141.4	6.0	10.4	33.2	1.0	148.9
Standard deviation	11.9	2.5	3.2	5.8	1.0	12.2
Average eccentricity	41.5	10.3	43.6	44.9	5.5	46.9

- In general, **as most number of nodes**, **more value in other Metrics.**
- **Brazil** x **USA**:
  - Brazil **more points,** more **average degree,** more **máx. degree** ;
  - but **USA more Variance.**

<< Alpha Version >>

Interactive Web Tools
“City Comparing”

Analysis with Econophysics Techniques

NormalScatters Visualizing | BinScatters Visualizing | ScatterGraph Visualizing | DataFrame Analysis

Main Dashboard: Comparing Cities, Countries and Macrorregions

Comparing Cities | Comparing Countries | Comparing Macrorregions

Choose 2 cities for compare

Deploy

Main Dashboard: Cities Compare

Choose 2 cities for compare

City 1:

Select Country

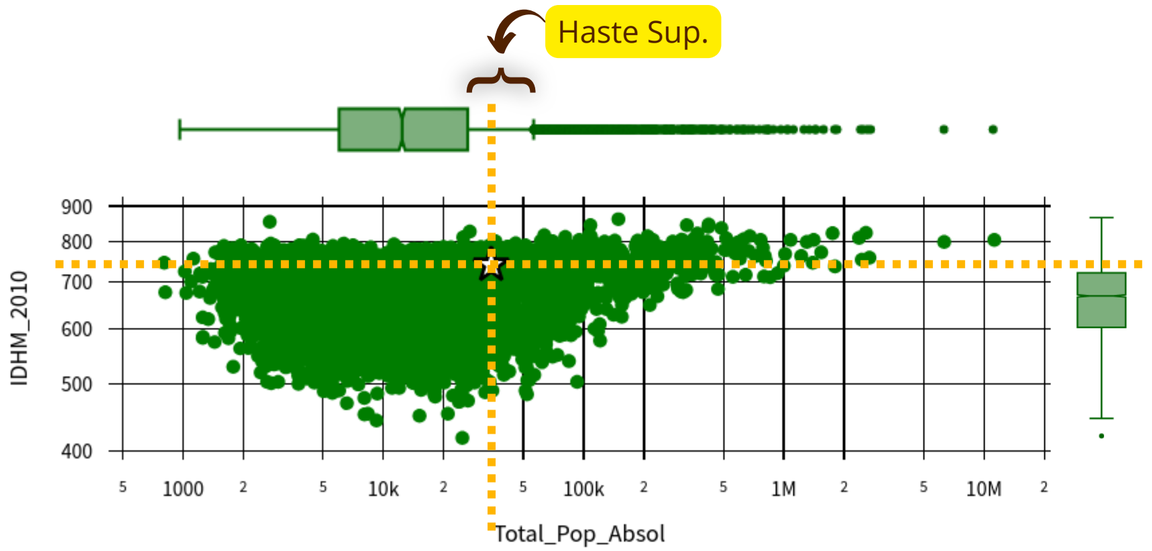
Brazil

Select State

RS

Select City

São Luiz Gonzaga (RS)



Legend

City 2:

Select Country

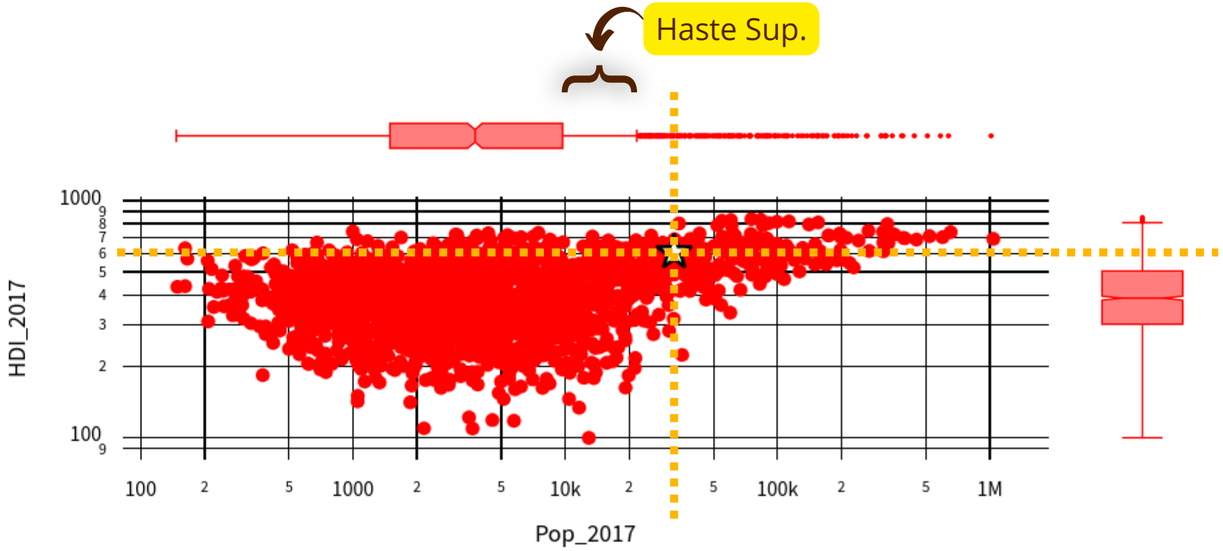
Peru

Select State

AMAZONAS

Select City

Chachapoyas



Legend

Conclusions

- It is a difficult challenge, but it is **possible to compare cities between countries**: especially if we use alternative **visual approaches**, with some numerical complements.
- The techniques developed and applied were able to be applied independently of scale and in different countries.
- B-GCA can be used as a graph clustering and colorization technique (ScatterGraph).
- **On average, the larger the population of the city, the higher the HDI of the respective city.** (Excluding the USA, in the sample of countries analyzed)

References:

1. A. Gomez Jonatan, B. Dipankar D., and C. Olfa N. A New Gravitational Clustering Algorithm (2003), DOI: https://doi.org/10.1137/1.9781611972733.8.
2. Suraj K. Sheth, Measuring and Luís M. A. Bettencourt. Health and human development in cities and neighborhoods in the United States (2023), DOI: https://doi.org/10.1038s42949-023-00088-y
3. Rodríguez Miranda, A., Vial Cossani, C., & Parrao, A. (2021). Índice Compuesto y Multidimensional de Desarrollo Regional: Una propuesta para América Latina. Revista Iberoamericana De Estudios Municipales, (23), 1–30. https://doi.org/10.32457/riem.v23i1.580

Goverment Data Sites (or oficial pdf informes HDI):

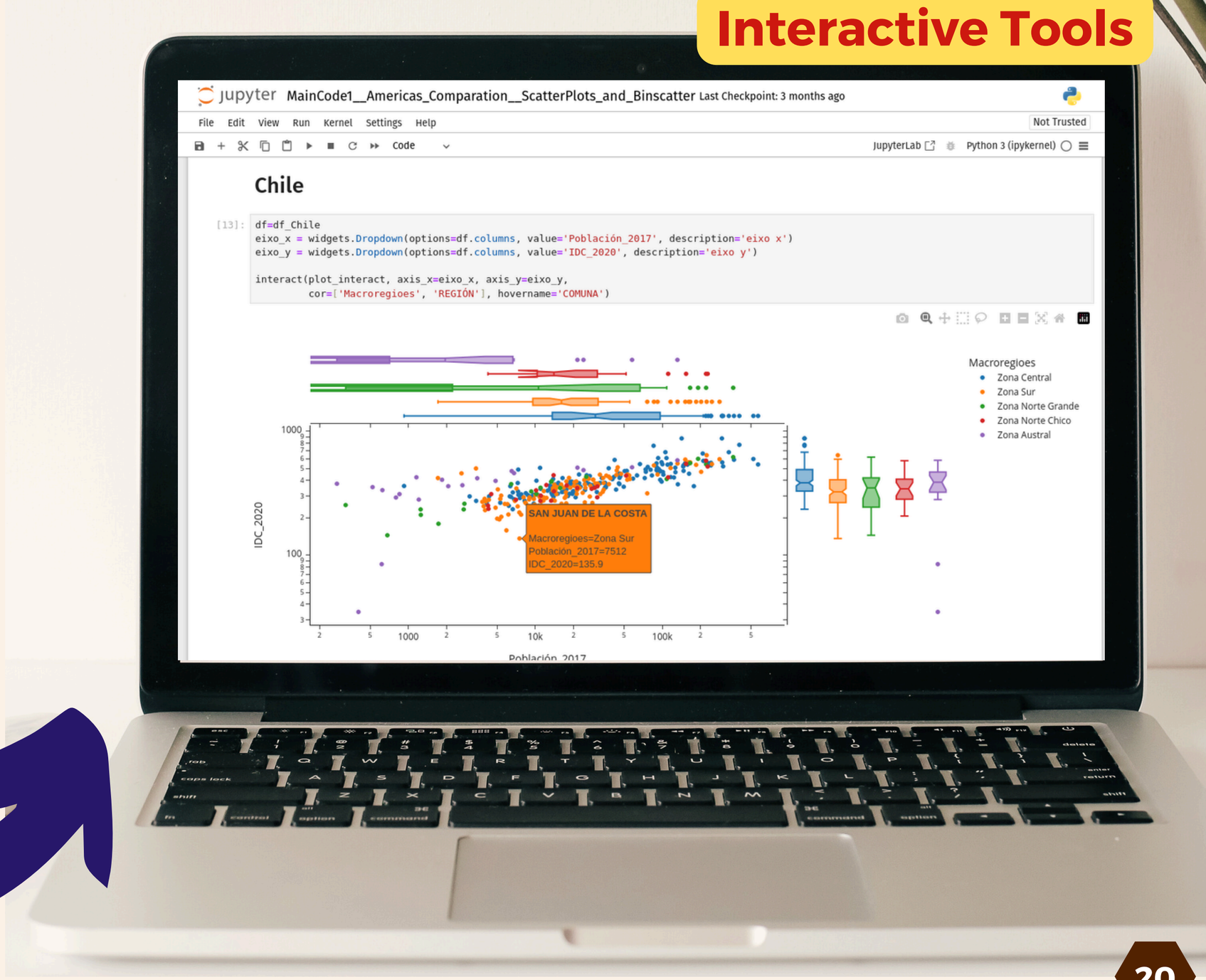
1. Mexico: https://www.inegi.org.mx/siscon/
2. Chile: https://repositorio.uautonoma.cl/bitstream/handle/20.500.12728/6742/V11_digital_final.pdf?sequence=1&isAllowed=y
3. Peru: https://www.up.edu.pe/egp/programas-especializacion_copy(1)/SiteAssets/Lists/Observatorio/AllItems/Informe%20de%20Evidencia%20IDH%202019.pdf
4. El Salvador: https://www.undp.org/sites/g/files/zskgke326/files/publications/El%20Salvador_MDGReport_2005_SP.pdf
5. Brazil: https://censo2010.ibge.gov.br/resultados.html
6. USA: https://web.archive.org/web/20110615084942/http://www.census.gov/hhes/www/poverty/poverty.html

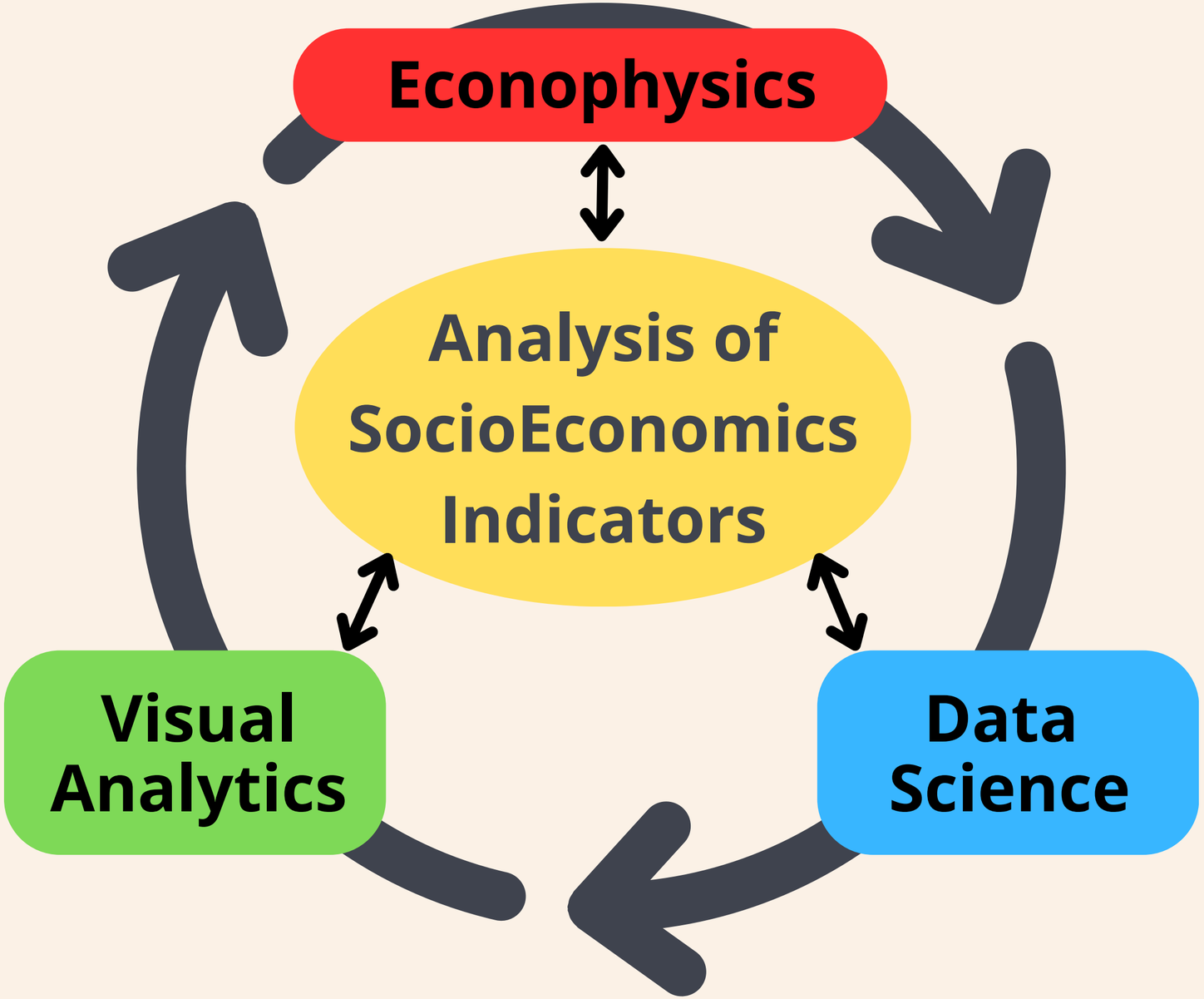
Thank you

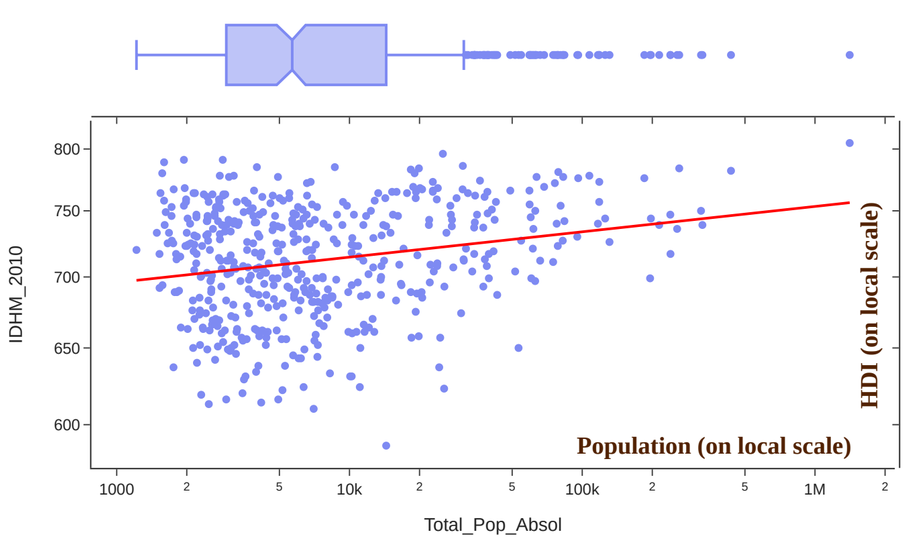
Github Paper

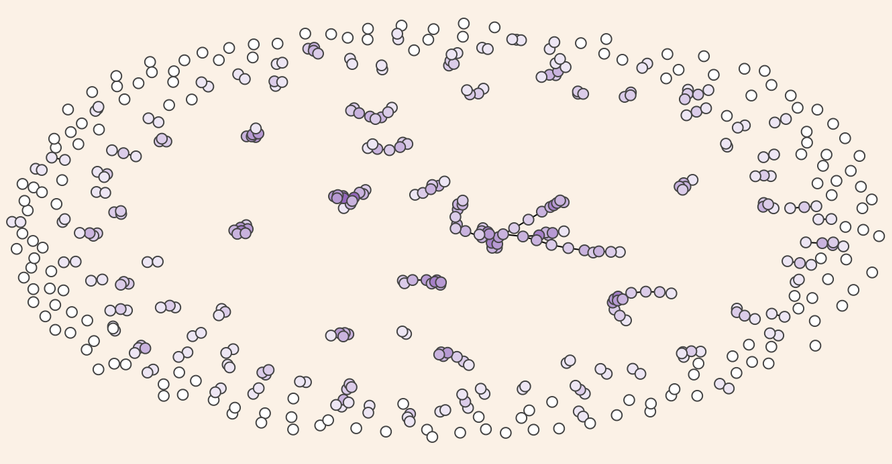
+ new works
+ new techniques

Interactive Tools

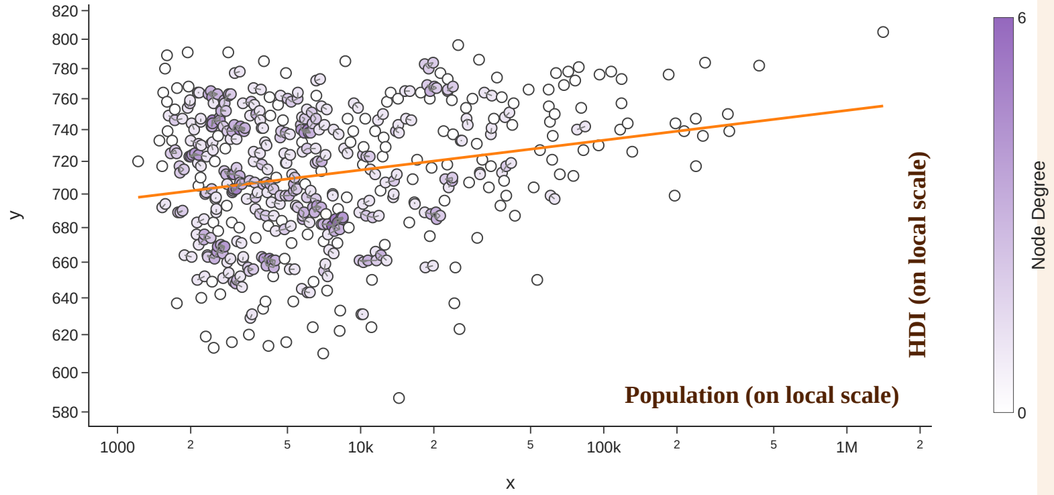



Econophysics
Analysis of SocioEconomics Indicators
Visual Analytics
Data Science

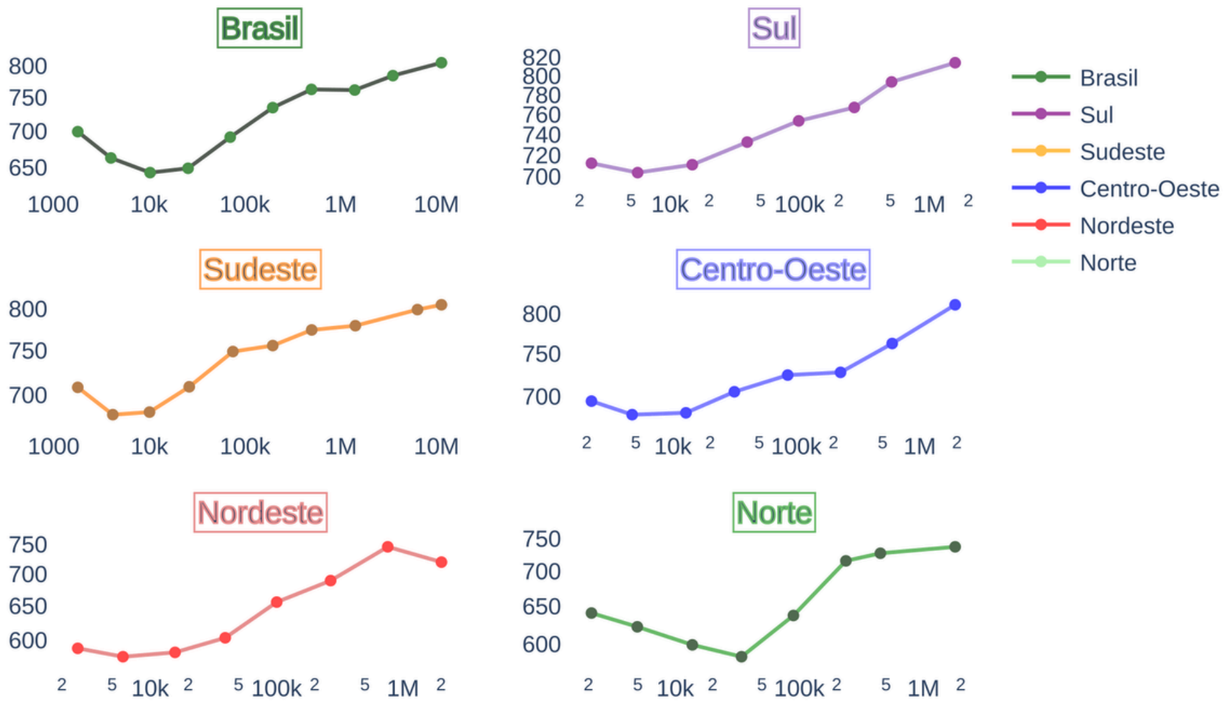


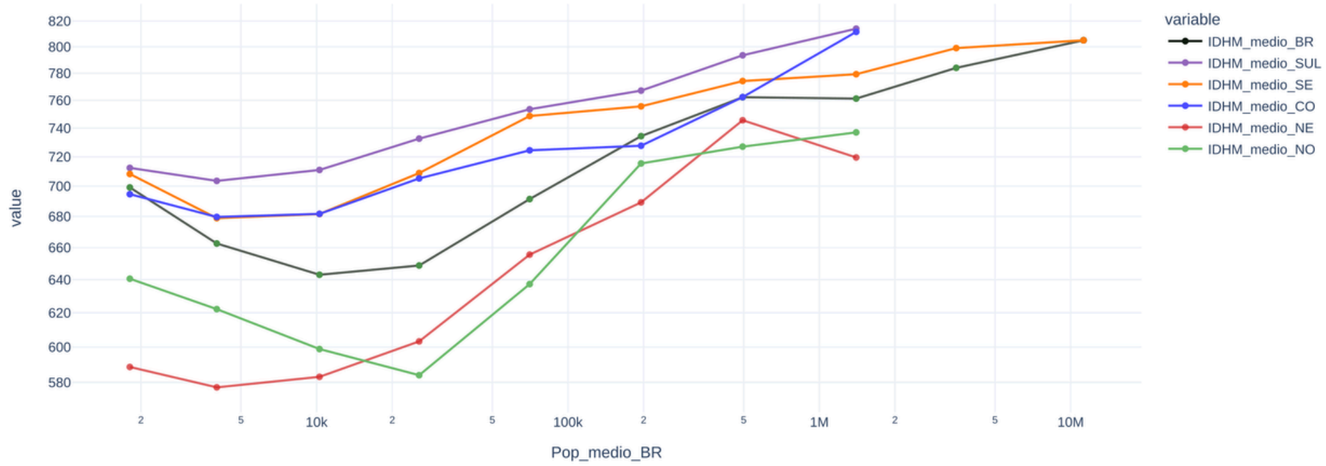


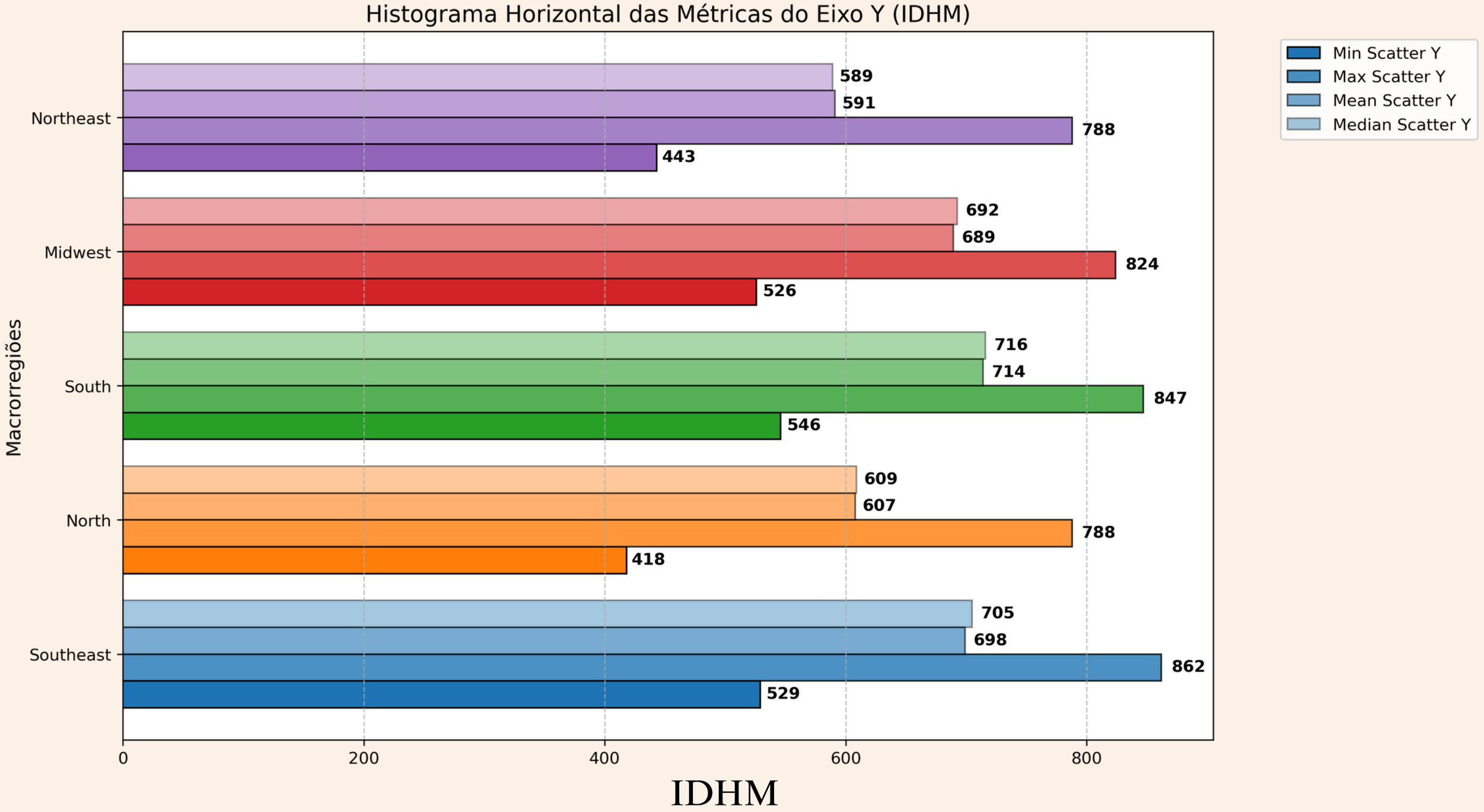
RS - Scatter Graph

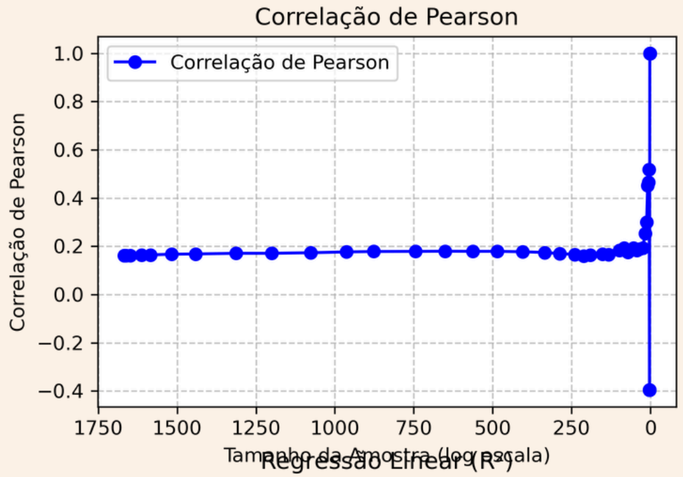


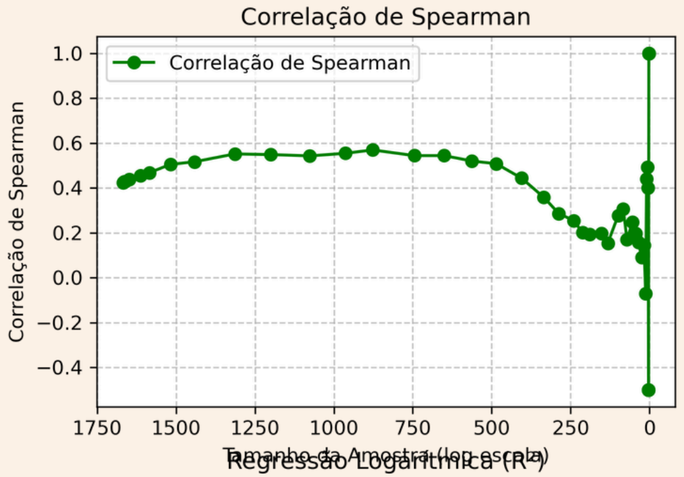
Análise IDHM x População Absoluta por bins

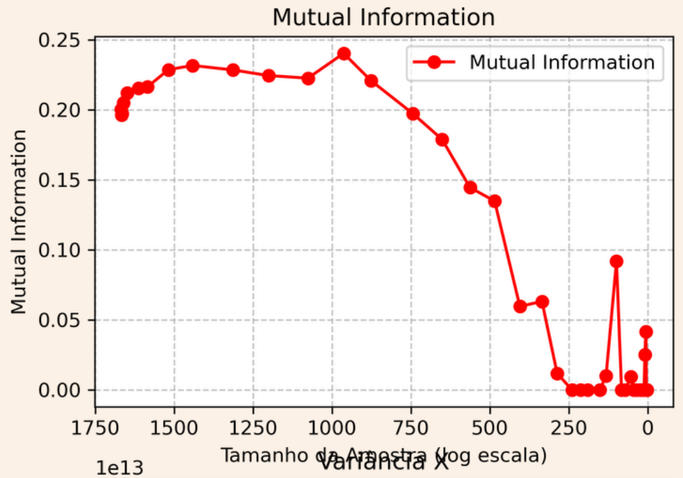


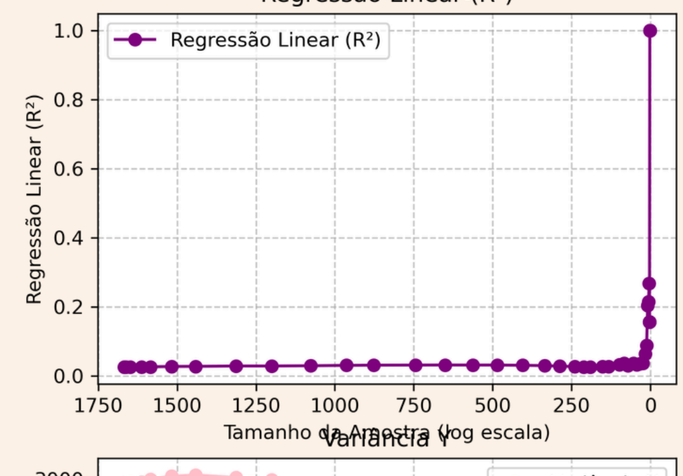


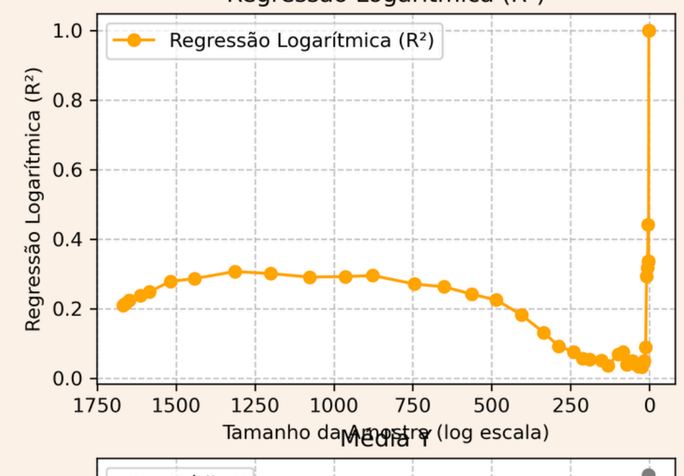

Histograma Horizontal das Métricas do Eixo Y (IDHM)
Min Scatter Y
Max Scatter Y
Mean Scatter Y
Median Scatter Y
Macrorregiões
Northeast
Midwest
South
North
Southeast
589
591
788
443
692
689
824
526
716
714
847
546
609
607
788
418
705
698
862
529
0
200
400
600
800
IDHM

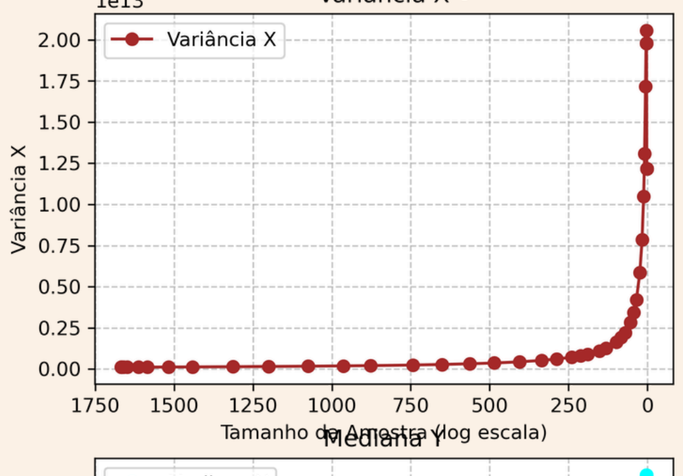

Correlação de Pearson
Correlação de Pearson
Correlação de Pearson
Tamanho da Amostra (log escala)

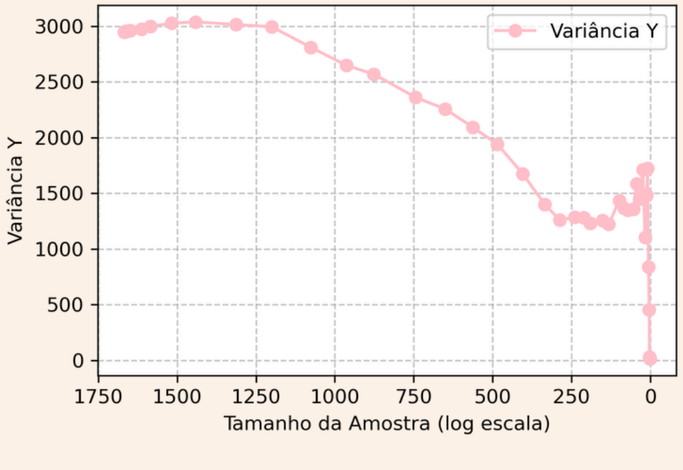

Correlação de Spearman
Correlação de Spearman
Correlação de Spearman
Tamanho da Amostra (log escala)

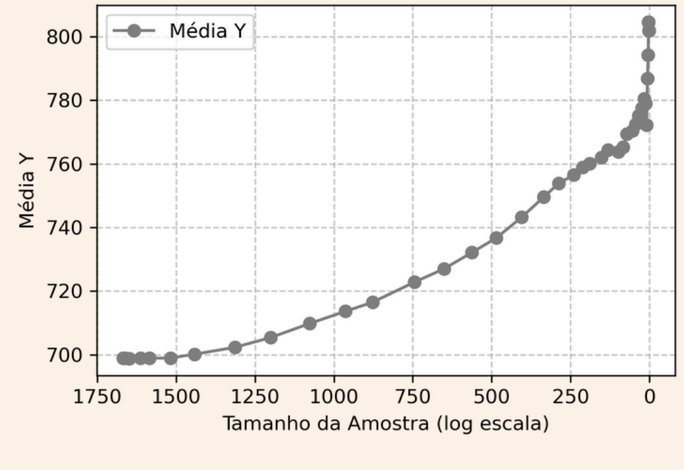

Mutual Information
Mutual Information
Mutual Information
Tamanho da Amostra (log escala)

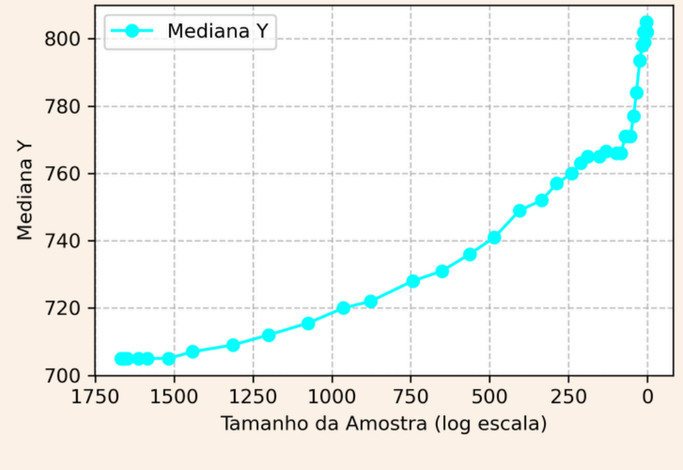

Regressão Linear (R²)
Regressão Linear (R²)
Regressão Linear (R²)
Tamanho da Amostra (log escala)


Regressão Logarítmica (R²)
Regressão Logarítmica (R²)
Regressão Logarítmica (R²)
Tamanho da Amostra (log escala)


Variância X
1e13
Variância X
Variância X
Tamanho da Amostra (log escala)


Variância Y
Variância Y
Variância Y
Tamanho da Amostra (log escala)


Média Y
Média Y
Média Y
Tamanho da Amostra (log escala)


Mediana Y
Mediana Y
Mediana Y
Tamanho da Amostra (log escala)